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Simulation of Power Demand of Machine Learning Workloads in German HPC Centers Using pandapower

Master's Thesis

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Abstract

The increasing use of machine learning (ML) workloads in high-performance computing (HPC) environments creates growing electrical power demand and raises questions about energy consumption, operational cost, and possible grid-related effects. This thesis develops a simulation framework for analyzing the power demand of ML workloads in HPC-like environments using pandapower.

The approach is based on measured single-GPU power traces from controlled training and inference runs. These traces are stored as CSV files and are used to derive time-dependent workload profiles. Since detailed real-world HPC infrastructure data was only partially available, the measured GPU traces are scaled using representative assumptions for non-GPU node power, node count, facility overhead, and number of simulated HPC centers. The resulting load profiles are evaluated in a synthetic grid model and in a SimBench-based benchmark grid.

The framework performs time-dependent power flow analysis and evaluates energy consumption, peak power demand, voltage behavior, transformer loading, line loading, power flow convergence, and operational cost. Several operational scenarios are considered, including GPU power limiting, PUE-based facility efficiency changes, workload shifting, load distribution across simulated HPC centers, capacity analysis, and comparison between synthetic and SimBench-based grid models.

The results show that different operational strategies affect the system in different ways. GPU power limiting reduces peak demand but can have a smaller effect on energy consumption when runtime increases. PUE-based facility efficiency reduces facility-level demand without changing workload runtime. Workload shifting can reduce operational cost without reducing physical energy consumption, while load distribution can reduce simultaneous peak demand without reducing same-work energy. The SimBench comparison shows that grid topology and existing network conditions strongly influence voltage and loading results.

Overall, the thesis demonstrates that measured ML workload power traces can be integrated into a reproducible power flow simulation framework. The results should be interpreted as scenario-based estimates rather than direct measurements of a real HPC facility, but the framework provides a useful basis for evaluating the interaction between ML workloads, facility-level power demand, and electrical grid models.

Zusammenfassung

Die zunehmende Nutzung von Machine-Learning-Workloads (ML) in High-Performance-Computing-Umgebungen (HPC) führt zu einem steigenden elektrischen Leistungsbedarf und wirft Fragen zu Energieverbrauch, Betriebskosten und möglichen netzbezogenen Auswirkungen auf. In dieser Arbeit wird ein Simulationsframework entwickelt, um den Leistungsbedarf von ML-Workloads in HPC-ähnlichen Umgebungen mit pandapower zu analysieren.

Der Ansatz basiert auf gemessenen Single-GPU-Leistungsprofilen aus kontrollierten Trainings- und Inferenzläufen. Diese Messdaten liegen als CSV-Dateien vor und werden zur Ableitung zeitabhängiger Workload-Profile verwendet. Da detaillierte reale HPC-Infrastrukturdaten nur teilweise verfügbar waren, werden die gemessenen GPU-Profile mithilfe repräsentativer Annahmen zu Nicht-GPU-Knotenleistung, Knotenzahl, Facility-Overhead und Anzahl simulierter HPC-Zentren skaliert. Die daraus entstehenden Lastprofile werden in einem synthetischen Netzmodell sowie in einem SimBench-basierten Benchmark-Netz ausgewertet.

Das Framework führt zeitabhängige Lastflussanalysen durch und bewertet Energieverbrauch, Spitzenleistungsbedarf, Spannungsverhalten, Transformatorauslastung, Leitungsauslastung, Lastflusskonvergenz und Betriebskosten. Dabei werden mehrere Betriebsszenarien untersucht, darunter GPU-Leistungsbegrenzung, PUE-basierte Änderungen der Facility-Effizienz, zeitliche Workload-Verschiebung, Lastverteilung auf simulierte HPC-Zentren, Kapazitätsanalyse sowie der Vergleich zwischen synthetischem und SimBench-basiertem Netzmodell.

Die Ergebnisse zeigen, dass verschiedene Betriebsstrategien unterschiedliche Auswirkungen haben. GPU-Leistungsbegrenzung reduziert die Spitzenlast, kann jedoch bei längerer Laufzeit einen geringeren Einfluss auf den Energieverbrauch haben. PUE-basierte Facility-Effizienz reduziert den Facility-Leistungsbedarf ohne Änderung der Workload-Laufzeit. Zeitliche Workload-Verschiebung kann Betriebskosten reduzieren, ohne den physikalischen Energieverbrauch zu senken, während Lastverteilung die gleichzeitige Spitzenlast reduzieren kann, ohne die Same-Work-Energie zu verringern. Der SimBench-Vergleich zeigt, dass Netztopologie und bestehende Netzbedingungen die Spannungs- und Auslastungsergebnisse deutlich beeinflussen.

Insgesamt zeigt die Arbeit, dass gemessene ML-Workload-Leistungsprofile in ein reproduzierbares Lastfluss-Simulationsframework integriert werden können. Die Ergebnisse sind als szenariobasierte Abschätzungen und nicht als direkte Messungen eines realen HPC-Standorts zu interpretieren. Dennoch bietet das Framework eine nützliche Grundlage zur Bewertung der Wechselwirkung zwischen ML-Workloads, Facility-Leistungsbedarf und

elektrischen Netzmodellen.

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List of Abbreviations

Abbreviation	Meaning
AC	Alternating Current
CPU	Central Processing Unit
CSV	Comma-Separated Values
DC	Direct Current
GPU	Graphics Processing Unit
HPC	High-Performance Computing
IT	Information Technology
ML	Machine Learning
MW	Megawatt
MWh	Megawatt-hour
PUE	Power Usage Effectiveness
pu	Per-unit
RAM	Random Access Memory

1. Introduction

1.1. Motivation

Machine learning (ML) has become an important driver of modern high-performance computing (HPC). Training and inference workloads increasingly rely on graphics processing units (GPUs) and other accelerators, because these devices provide the parallel computing performance required for data-intensive models. As a result, ML workloads are not only relevant from a software and hardware performance perspective, but also from an electrical power demand perspective.

The electricity demand of data centers has received growing attention in recent years. According to the International Energy Agency, data centers and data transmission networks already account for a relevant share of global electricity use, and demand is expected to increase further with the expansion of AI-related infrastructure [4], [5]. HPC centers are especially relevant in this context because they operate computationally intensive workloads that can create high and time-dependent power demand. ML training runs may keep GPUs active for extended periods, while inference workloads can show variable utilization depending on request rates, batching strategies, latency requirements, and deployment configuration.

For operators of HPC systems, this development creates several practical challenges. Increasing power demand affects operational cost, especially when electricity prices vary over time. In addition, the total facility power is not determined only by IT equipment. Cooling, power distribution, and auxiliary infrastructure also contribute to the total energy demand. This relationship is commonly described using Power Usage Effectiveness (PUE), which relates total facility energy to IT equipment energy [8]. A lower PUE indicates lower non-IT overhead for the same IT load. Furthermore, high peak loads may become relevant for the electrical infrastructure, especially when many compute nodes or several HPC centers operate simultaneously.

From a power system perspective, ML workloads are interesting because their electrical demand is not constant. GPU power changes during execution depending on workload phase, utilization, batch size, model behavior, and hardware configuration. Therefore, a single nominal power value or average power estimate is not sufficient to evaluate how such workloads affect an electrical grid. A time-series representation is needed to analyze peak power, accumulated energy, voltage behavior, transformer loading, and line loading.

At the same time, detailed real-world HPC infrastructure data is often difficult to obtain. Power measurements from computing centers may be incomplete, aggregated, confidential, or not directly linked to individual workloads. This makes it difficult to build a fully site-specific model of an HPC facility. A simulation-based approach offers a practical alternative: measured workload power traces can be scaled into infrastructure-level load profiles and evaluated in controlled grid models.

This thesis is motivated by the need to connect workload-level ML power behavior with grid-level power system analysis. By combining measured single-GPU workload traces, infrastructure-level scaling assumptions, and pandapower-based power flow simulation, the work provides a structured way to evaluate energy consumption, peak demand, operational cost, and grid-related effects in HPC-like environments.

1.2. Problem Statement

ML workloads in HPC environments create dynamic electrical load profiles. However, these profiles are difficult to evaluate using static assumptions such as nominal GPU power or simple average power. Such values can be useful for rough estimates, but they do not capture temporal variation during workload execution. Training and inference runs can contain phases with different utilization levels, short power peaks, idle periods, and changing runtime behavior. These effects become important when the workload is analyzed not only as a computational task, but also as an electrical load.

A central problem is the missing connection between workload-level measurements and grid-level evaluation. GPU power traces describe the electrical behavior of a workload over time, but they do not directly show how the resulting demand affects voltage levels, transformer loading, line loading, or power flow feasibility. Conversely, power flow tools can evaluate electrical networks, but they require suitable load profiles as input. This creates a modeling gap between ML workload measurements and power system simulation.

This thesis addresses this gap by deriving time-dependent HPC-center load profiles from measured single-GPU power traces and additional system-level assumptions. These assumptions include non-GPU node power, the number of nodes per center, the number of simulated centers, and facility overhead represented through PUE. The resulting load profiles are assigned to controllable loads in pandapower and evaluated using AC power flow simulations.

A second problem is the limited availability of detailed real-world HPC infrastructure data. In practice, power measurements from HPC facilities are often aggregated or not directly mapped to specific ML jobs. Detailed information about internal electrical infrastructure may also be unavailable. Therefore, this thesis does not attempt to reproduce one specific HPC site. Instead, it uses two reproducible grid representations: a synthetic grid model based on representative German voltage levels and a SimBench-based benchmark grid.

The evaluation must also go beyond total energy consumption. A workload with the same energy demand can have different operational effects depending on when it is executed, how high its peak power is, how efficient the facility overhead is, and whether multiple centers operate at the same time. Therefore, the thesis evaluates energy consumption, peak demand, operational cost, voltage behavior, transformer loading, line loading, and power flow convergence together.

The main problem can be summarized as follows: ML workloads create time-dependent electrical load profiles, but their effects on energy consumption, operational cost, and grid behavior are difficult to evaluate without a combined workload modeling and power flow simulation approach.

1.3. Thesis Aim and Contributions

The aim of this thesis is to develop and evaluate a simulation framework that connects ML workload power traces with power system analysis. The framework converts measured single-GPU power traces into time-dependent HPC-center load profiles, scales them using configurable infrastructure assumptions, and evaluates them in synthetic and benchmark-based grid models.

Research objectives. The thesis follows six research objectives:

1. To process measured single-GPU power traces from controlled ML workload executions and convert them into time-series load profiles suitable for electrical simulation.
2. To model simulated HPC centers as aggregated electrical loads by including non-GPU node power, node count, center count, and facility overhead represented through PUE.
3. To integrate the resulting load profiles into pandapower-based AC power flow simulations using both a synthetic grid model and a SimBench-based benchmark grid.
4. To evaluate energy consumption, peak power demand, voltage behavior, transformer loading, line loading, power flow convergence, and operational cost.
5. To compare operational strategies such as GPU power limiting, workload shifting, PUE-based facility efficiency changes, and load distribution across simulated HPC centers.
6. To assess the usefulness and limitations of the proposed approach when detailed real-world HPC infrastructure data is only partially available.

Contributions. The main contribution of this thesis is a reproducible simulation pipeline that links measured ML workload power traces with AC power flow analysis. The pipeline processes workload measurements, derives time-dependent load profiles, scales them to simulated HPC-center level, and evaluates their grid impact in pandapower.

A second contribution is the use of two complementary grid modeling approaches. The synthetic grid model provides a transparent and controllable environment for scenario analysis, while the SimBench-based grid provides a standardized benchmark network for comparison.

A third contribution is the integration of energy and cost evaluation into the simulation workflow. Besides technical quantities such as peak power, voltage, transformer loading, and line loading, the framework also estimates operational cost using time-dependent electricity prices.

Finally, the thesis evaluates several operational strategies under the same baseline assumptions. This makes it possible to compare how GPU power limiting, PUE-based efficiency changes, workload shifting, and load distribution affect energy demand, peak power, cost, and grid-related metrics.

1.4. Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews related work on data center energy demand, HPC energy efficiency, ML workload power behavior, data center efficiency metrics, and power system simulation. Chapter 3 introduces the technical background required for the thesis, including HPC systems, ML workloads, GPU power behavior, PUE, and AC power flow analysis.

Chapter 4 describes the data and workload modeling approach. It explains how measured GPU power traces are preprocessed and converted into node-level, center-level, and multi-center load profiles. Chapter 5 presents the simulation methodology, including the synthetic grid model, the SimBench-based grid model, the assignment of HPC loads, the time-series power flow procedure, and the evaluated operational scenarios.

Chapter 6 explains the implementation of the Python-based simulation framework and the Streamlit dashboard. Chapter 7 presents the simulation results and evaluates the baseline case, operational scenarios, capacity analysis, and SimBench comparison. Chapter 8 discusses the interpretation, usefulness, and limitations of the approach. Chapter 9 concludes the thesis and outlines possible directions for future work.

2. Related Work

This chapter reviews work related to data center and HPC energy demand, energy-aware workload management, ML workload power behavior, data center efficiency metrics, and power system simulation. The purpose is to position this thesis within existing research and to identify the gap addressed by the proposed simulation framework.

2.1. Energy Demand of Data Centers and HPC Systems

The electricity demand of data centers has been studied for many years, as computing infrastructure has become a central part of modern digital services. Early work by Koomey showed that data center electricity use had already become a relevant part of global electricity consumption and that its growth depended on server deployment, utilization, and infrastructure efficiency **koomey2011datacenter**. More recent reports by the International Energy Agency indicate that the expansion of cloud computing, artificial intelligence, and data-intensive applications continues to increase the importance of data center electricity demand [4], [5].

Data center energy demand is not determined only by the installed IT equipment. It also depends on utilization, cooling systems, power distribution, redundancy requirements, and operational management. Surveys on data center energy modeling show that energy consumption can be modeled at different levels, including component level, server level, rack level, and facility level [2]. This multi-level view is relevant for this thesis because the available measurements describe workload-level GPU power, while the simulation requires an estimate of center-level facility power.

HPC systems represent a specific case within the broader data center domain. Unlike many general-purpose cloud or enterprise data centers, HPC centers are designed for computationally intensive workloads and often operate large numbers of CPUs, GPUs, accelerators, high-performance interconnects, and shared storage systems. Their electrical demand can therefore be high and strongly dependent on the workload being executed. In the context of ML workloads, this becomes especially relevant because training and inference tasks can keep accelerators active for extended periods or create time-varying utilization patterns.

Energy efficiency in HPC has been studied from several perspectives, including hardware efficiency, power-aware job scheduling, workload placement, and resource management **eehpc_survey**, **kocot2023energy**. These works show that energy-aware operation is an important research topic in HPC. However, many of these approaches focus mainly on the computing system itself. They usually evaluate energy consumption, performance, utilization, or scheduling objectives, while the direct connection between workload-level power traces and grid-level power flow behavior is less central.

This thesis builds on this research direction but shifts the evaluation perspective. Instead of considering only the energy consumed by the computing system, it also evaluates how the workload-derived load profile behaves when applied to an electrical grid model. This makes it possible to analyze not only energy consumption, but also peak demand, voltage behavior, transformer loading, line loading, and operational cost.

2.2. Energy-Aware Scheduling and Resource Management

Energy-aware scheduling and resource management are important methods for reducing the power and energy impact of large computing systems. In cloud and data center environments, energy-aware resource allocation has often focused on consolidating workloads, improving utilization, and reducing the number of underutilized active servers [1]. In HPC environments, the problem is more constrained because workloads often have strict performance requirements, parallel execution constraints, queueing policies, and dependencies between jobs.

Surveys on HPC energy-aware scheduling describe several optimization goals, including minimizing total energy, limiting peak power, satisfying power caps, reducing energy-delay product, and balancing performance with energy consumption **kocot2023energy**. These goals are closely related to the operational scenarios considered in this thesis. For example, GPU power limiting is related to power capping, workload shifting is related to time-aware scheduling, and load distribution is related to reducing simultaneous peak demand.

However, there is an important distinction between energy reduction and peak reduction. A scheduling strategy can reduce the instantaneous power demand seen by the grid without reducing the total amount of work that must be executed. If the same workload is delayed or distributed over a longer time interval, the peak load may decrease while the same-work energy remains unchanged. This distinction is important for interpreting the load distribution scenario in this thesis.

Existing scheduling studies provide important foundations for understanding how workloads can be managed from a computing-system perspective. The contribution of this thesis is different: the scheduling and operational strategies are evaluated through a workload-to-grid simulation pipeline. This allows their impact on electrical metrics such as voltage and component loading to be analyzed in addition to power, energy, and cost.

2.3. Power Demand of Machine Learning Workloads

Machine learning workloads differ from many traditional computing workloads because they often rely strongly on GPUs or other accelerators. During training, GPUs can operate at high utilization for extended periods, especially when large models, large datasets, and high batch sizes are used. Previous work has shown that training modern deep learning models can require substantial computational resources and energy, making energy demand an important factor in the design and operation of ML systems **strubell2019energy**, **patterson2021carbon**.

A related issue is the reporting of energy and carbon impacts in ML research. Henderson et al. argue that systematic reporting of energy use and carbon emissions is important for understanding the environmental impact of ML experiments [3]. This is relevant for this thesis because it supports the use of measured power traces instead of relying only on nominal hardware specifications. Direct or trace-based power information provides a more detailed basis for energy estimation than static hardware ratings.

Inference workloads can show different power behavior from training workloads. While training is often executed as a longer batch job, inference systems may react to changing request rates, batching strategies, latency requirements, and deployment configuration. Dynamic batching can combine multiple requests into a single batch to improve throughput and resource utilization [7]. As a result, inference demand is not determined only by the model and hardware, but also by how requests arrive and how the serving system schedules them.

For this thesis, the important methodological point is that ML workload power demand should not be represented only by a single nominal GPU power value. Nominal power or thermal design power can be useful for rough estimates, but it hides temporal variation during workload execution. Since grid behavior depends on instantaneous load, time-dependent power traces are more suitable for evaluating peak demand, accumulated energy, voltage behavior, transformer loading, and line loading.

In this work, GPU power traces are therefore used as the starting point for workload modeling. These traces are not used directly as complete grid loads. Instead, they are transformed into simulated HPC-center load profiles by adding non-GPU node power assumptions, scaling parameters, and facility overhead represented through PUE. This allows workload-level measurements to be evaluated at a higher infrastructure level.

2.4. Data Center Efficiency and PUE

The electrical demand of a data center is higher than the demand of the IT equipment alone. In addition to servers, GPUs, storage, and network hardware, supporting infrastructure such as cooling systems, power distribution equipment, fans, pumps, and auxiliary systems also contributes to total facility power demand. For this reason, data center efficiency is often discussed using Power Usage Effectiveness (PUE), which relates total facility energy demand to IT equipment energy demand [8].

PUE is commonly defined as

$$\mathrm{PUE} = \frac{E_{\text{facility}}}{E_{\text{IT}}}, \tag{2.1}$$

where E_{facility} denotes total facility energy and E_{IT} denotes IT equipment energy. A PUE value close to 1 indicates that only a small amount of additional energy is required beyond the IT load, whereas higher values indicate larger facility overhead.

In this thesis, PUE is used as an aggregated modeling parameter. It does not describe the individual behavior of cooling units, uninterruptible power supplies, airflow, pumps, or power distribution losses in detail. Instead, it scales the modeled IT power demand to estimate facility-level power demand. This is consistent with the scope of the thesis, which does not attempt to model the thermodynamic behavior of a specific data center.

The distinction between PUE-based efficiency changes and workload-level power management is important. Reducing PUE decreases facility power for the same IT load, but it does not directly change the computational runtime of the workload. In contrast, GPU power limiting can reduce instantaneous GPU power while potentially increasing runtime. Therefore, PUE-based scenarios and GPU power limiting scenarios must be interpreted differently in the evaluation.

2.5. Power System Simulation and Benchmark Grids

Power flow analysis is widely used to evaluate the operating state of electrical networks under given load and generation conditions. In a power flow calculation, the network topology, line parameters, transformer parameters, generation units, external grid connection, and electrical loads are used to calculate bus voltages, line loading, transformer loading, and power flows. These quantities are relevant for this thesis because the objective is not only to estimate the energy demand of ML workloads, but also to evaluate how workload-derived loads affect grid-related behavior.

This thesis uses pandapower for electrical network simulation. Pandapower is an open-source Python-based tool for modeling, analyzing, and optimizing electrical power systems [9]. It provides data structures for buses, lines, transformers, loads, generators, and external grid connections, and supports AC power flow calculations. This makes it suitable for integrating workload-derived load profiles into a power system model and evaluating the resulting electrical state over time.

When real grid data is not available, benchmark grids provide reproducible reference models. SimBench is one such benchmark dataset and provides standardized power system networks for different voltage levels and study cases [6]. In this thesis, SimBench is used as a benchmark-based grid source. The SimBench-based model is not interpreted as a real HPC site, but as a standardized network into which simulated HPC loads can be inserted.

The use of both a synthetic grid and a SimBench-based grid supports the methodology of this thesis. The synthetic grid provides a transparent and controllable environment with explicit HPC load connection points. The SimBench-based grid provides a standardized benchmark structure. Together, these models allow the workload-to-grid pipeline to be evaluated under both controlled and benchmark-based assumptions.

2.6. Research Gap

The reviewed literature shows that data center energy demand, HPC energy-aware scheduling, ML workload energy consumption, data center efficiency metrics, and power system simulation are established research areas. However, these topics are often studied separately. Work on HPC and ML energy use usually focuses on the computing infrastructure, workload execution, hardware utilization, or scheduling objectives. In contrast, power system studies usually start from predefined electrical load profiles and focus on grid operation.

For this thesis, the relevant gap lies in the connection between these perspectives. Existing work provides foundations for energy-aware computing and power system analysis, but the complete chain from measured ML workload power behavior to infrastructure-level demand and grid-level power flow evaluation is less commonly addressed. This is important because average energy consumption alone does not show how time-dependent workload demand affects peak load, voltage behavior, component loading, or operational cost.

This thesis addresses the gap by combining workload trace processing, infrastructure-level scaling, facility power modeling, AC power flow simulation, and operational cost estimation in one reproducible framework. The approach does not claim to reproduce a specific real HPC site. Instead, it provides a scenario-based method for evaluating how workload-derived power profiles behave under different infrastructure, scheduling, and grid assumptions.

3. Background

This chapter introduces the technical concepts required for the simulation framework developed in this thesis. It defines the main terminology used in the following chapters, including HPC systems, ML workloads, GPU power behavior, facility-level power demand, and AC power flow analysis. The purpose of this chapter is to provide the technical basis for the modeling and simulation approach, while the detailed simulation methodology is described later in Chapter 5.

3.1. High-Performance Computing Systems

High-performance computing (HPC) systems are designed to execute computationally demanding workloads that exceed the capabilities of conventional desktop or small server systems. Such systems typically consist of many interconnected compute nodes, shared storage systems, high-performance networks, and centralized management infrastructure. Depending on the system design, compute nodes may contain CPUs, GPUs, or other accelerators.

In this thesis, an HPC center is not modeled at the level of individual users, scheduler queues, or specific jobs. Instead, it is represented as an aggregated electrical load. The power demand of this load is derived from measured workload traces and scaling assumptions. This abstraction is appropriate because the focus of the thesis is not the internal scheduling behavior of a real HPC cluster, but the resulting electrical demand seen by a grid model.

A simplified view of the modeled power chain is shown in Figure 3.1. The workload-level GPU trace is first converted into a node-level power estimate. This estimate is then scaled to the level of a simulated HPC center and finally represented as a time-dependent electrical load in the grid model.

modeling_chain.png

Figure 3.1.: Simplified modeling chain from ML workload power traces to HPC-center-level electrical load.

3.2. Machine Learning Workloads

Machine learning workloads describe computational tasks in which models learn patterns from data or apply learned models to new inputs. In the context of this thesis, two workload types are relevant: training and inference. Training refers to the process of adjusting model parameters using data, while inference refers to using an already trained model to produce predictions or outputs.

Training workloads are usually compute-intensive and can run for extended periods, especially when large models, large datasets, or many training iterations are involved. These workloads often use GPUs because many deep learning operations, such as matrix multiplications and convolutions, can be parallelized efficiently. As a result, training can lead to sustained GPU utilization and significant power demand.

Inference workloads can have different power characteristics. A single inference request may require less computation than a full training process, but deployed inference systems may process many requests over time. Their utilization depends on factors such as request rate, batching strategy, latency requirements, model size, and deployment configuration. Therefore, inference demand may be more variable than a long training run, especially in service-oriented environments.

For power demand simulation, the distinction between training and inference is important because both workload types can create different temporal load profiles. A training run may produce a relatively sustained power profile, while inference may create more variable demand depending on how requests are served. In both cases, a time-dependent representation is more informative than a single static power value.

In this thesis, ML workloads are represented through single-GPU power traces stored as CSV files. These traces provide time-dependent GPU power and utilization measurements from controlled workload executions. They are not treated as complete data-center measurements. Instead, they are used as workload-level input signals that are scaled to node, center, and multi-center level using the modeling approach described in Chapter 4.

3.3. GPU Power Behavior

GPUs are central to many modern ML workloads because they can execute a large number of parallel operations efficiently. Their power consumption is not constant during execution. Instead, it depends on utilization, memory activity, workload phase, batch size, clock behavior, and power management settings. For this reason, the nominal power rating of a GPU is not sufficient to describe the actual power behavior of a workload over time.

In the available workload traces, GPU power is recorded as a time-dependent value in watts. This allows the workload to be represented as a power signal rather than as a single static number. Such a representation is important because short peaks, changes in utilization, and idle periods can influence the resulting load profile after scaling.

The measured GPU power values describe the behavior of the GPU during a controlled run, but they do not include the complete power demand of an HPC node or data center. A full compute node also includes CPU power, memory, storage, network components, and supporting electronics. At facility level, additional cooling and power distribution overheads must also be considered.

In the simulation framework, the GPU power trace forms the dynamic part of the load model. Additional assumptions are then added for non-GPU node power, node count, facility overhead, and the number of simulated centers. This preserves the temporal behavior of the measured workload while making it possible to evaluate larger HPC-like scenarios.

3.4. Facility Power and PUE

The electrical demand of an HPC facility is higher than the direct IT load. Besides GPUs, CPUs, memory, storage, and networking hardware, supporting infrastructure is required to operate the facility. This includes cooling systems, power distribution equipment, uninterruptible power supply systems, fans, pumps, and auxiliary components.

As introduced in Section 2.1, Power Usage Effectiveness (PUE) relates total facility energy demand to IT equipment energy demand. In this thesis, PUE is used as an aggregated scaling parameter. It does not model individual cooling units, power conversion stages, airflow behavior, or thermal dynamics. Instead, it converts the modeled IT power into an estimated facility-level power demand.

The distinction between IT power and facility power is important for scenario evaluation. A reduction in PUE lowers the total facility power for the same IT workload, but it does not directly change the computational runtime of the workload. This differs from GPU power limiting, where a lower GPU power level may also influence execution time. Therefore, PUE-based facility efficiency scenarios and workload-level power management scenarios must be interpreted separately.

3.5. Power Flow Analysis

Power flow analysis is used to determine the steady-state operating condition of an electrical network for a given set of loads, generation units, and network parameters. It calculates electrical quantities such as bus voltage magnitudes, active and reactive power flows, line loading, and transformer loading. These quantities are important for this thesis because the goal is not only to estimate workload energy consumption, but also to evaluate how the resulting load affects the electrical grid model.

In AC power flow analysis, both active power and reactive power are considered. Active power represents the useful power consumed by electrical loads, while reactive power is related to energy exchange in the electric and magnetic fields of AC systems. In the simulation framework, the workload-derived demand is primarily modeled as active power. Reactive power is estimated using a simplified proportional assumption so that the loads can be represented in a form suitable for AC power flow calculations.

The main grid-related outputs used in this thesis are voltage magnitude, transformer loading, line loading, and convergence status. Voltage magnitude is evaluated in per-unit values, which allows voltages at different nominal voltage levels to be compared in normalized form. Transformer and line loading are evaluated as percentages of their rated capacity. These metrics indicate whether the simulated workload demand causes noticeable stress in the modeled electrical network.

The power flow calculations in this thesis are performed using pandapower [9]. For each time step of the workload profile, the corresponding load values are assigned to the grid model and an AC power flow calculation is executed. The resulting time series of voltage levels, transformer loading, and line loading is then used to evaluate the effect of different workload and infrastructure scenarios.

4. Data and Workload Modeling

This chapter describes the workload input data and explains how measured GPU power traces are transformed into time-dependent load profiles for the power system simulation. The purpose is to define the modeling chain from measured workload-level power to node-level, center-level, and multi-center electrical demand. The resulting load profiles are used as input for the simulation methodology described in Chapter 5.

4.1. Input Data

The workload input used in this thesis consists of CSV-based GPU power traces from controlled ML workload executions. The available data contains training and inference traces. Each trace contains time-dependent measurements such as timestamps, GPU utilization, GPU power, total GPU power, CPU utilization, and the number of GPUs used during the run.

The available traces represent single-GPU workload executions. They are therefore not interpreted as complete HPC-center or data-center power measurements. Instead, they are treated as workload-level input signals that describe how GPU power changes during execution. These signals are then extended with additional assumptions in order to estimate node-level, center-level, and multi-center electrical demand.

The most important input variable for the load modeling process is the measured GPU power. In the CSV traces, this value is given in watts. The GPU power trace provides the dynamic part of the model, while non-GPU node power, number of nodes, PUE, and number of simulated centers are added as configurable modeling assumptions.

4.2. Trace Preprocessing

Before the workload traces can be used in the simulation, they are converted into a consistent time-series format. The preprocessing step loads the CSV files, checks the required columns, converts timestamps, sorts the data chronologically, and calculates the time difference between consecutive samples.

For a single run, the measured GPU power trace is represented as

$$P_{\mathrm{GPU}}(t_1), P_{\mathrm{GPU}}(t_2), \ldots, P_{\mathrm{GPU}}(t_n), \tag{4.1}$$

where $P_{\mathrm{GPU}}(t_i)$ denotes the measured GPU power at time step t_i. The time interval between two consecutive samples is calculated as

$$\Delta t_i = t_i - t_{i-1}. \tag{4.2}$$

In the implementation, this interval is stored as `delta_seconds`. If invalid or missing time intervals occur, the implementation uses a fallback value based on the median sampling interval of the trace. This prevents individual timestamp irregularities from breaking the energy calculation.

When multiple runs are available, the traces are aligned to a common length and averaged into a representative workload profile. For m available runs, the representative GPU power at time step t_i is calculated as

$$\overline{P}_{\mathrm{GPU}}(t_i) = \frac{1}{m} \sum_{j=1}^{m} P_{\mathrm{GPU},j}(t_i), \tag{4.3}$$

where $P_{\mathrm{GPU},j}(t_i)$ is the GPU power of run j at time step t_i. This averaging step reduces the influence of individual run variation and produces one representative time-dependent workload profile for the later scaling steps.

4.3. Energy Calculation

Energy consumption is calculated by integrating the sampled power values over time. Since the workload trace is discrete, the energy is computed as a sum over all samples. If the time interval is given in seconds, it is first converted to hours:

$$\Delta t_{h,i} = \frac{\Delta t_i}{3600}. \tag{4.4}$$

If power is given in watts, the trace energy in watt-hours is calculated as

$$E_{\mathrm{Wh}} = \sum_{i=1}^{n} P_{\mathrm{W}}(t_i) \cdot \Delta t_{h,i}. \tag{4.5}$$

For facility-level results, the same principle is applied to the simulated facility power. If the power is given in megawatts, the energy in megawatt-hours is calculated as

$$E_{\mathrm{MWh}} = \sum_{i=1}^{n} P_{\mathrm{MW}}(t_i) \cdot \Delta t_{h,i}. \tag{4.6}$$

This calculation is used throughout the evaluation to compare baseline and scenario results. It is important that the energy calculation uses the actual sampling intervals from the trace, because the workload duration is not assumed to be exactly one hour.

4.4. From GPU Power to Node Power

A GPU power trace only describes the power behavior of the GPU during workload execution. An HPC compute node also consumes power through other components, including the CPU, memory, storage, network interface, and supporting electronics. Therefore, the GPU power trace is extended with assumptions for non-GPU node power.

The node-level power is modeled as

$$P_{\mathrm{node}}(t) = P_{\mathrm{GPU}}(t) + P_{\mathrm{CPU}} + P_{\mathrm{RAM}} + P_{\mathrm{storage}} + P_{\mathrm{network}}. \tag{4.7}$$

Here, $P_{\mathrm{GPU}}(t)$ is time-dependent, while the remaining terms are modeled as constant assumptions in the current implementation. This preserves the temporal behavior of the measured GPU trace while adding a simplified estimate for the rest of the compute node.

This modeling choice is a simplification. In a real system, CPU, memory, storage, and network power can also vary over time. However, the available measurement data focuses on GPU power, and detailed node-level component measurements were not available. The constant non-GPU terms therefore provide a transparent and configurable approximation.

4.5. Scaling to HPC-Center Load

After calculating node-level power, the load is scaled to represent one simulated HPC center. If N_{nodes} denotes the number of nodes per center, the IT power of one simulated center is modeled as

$$P_{\mathrm{IT,center}}(t) = N_{\mathrm{nodes}} \cdot P_{\mathrm{node}}(t). \tag{4.8}$$

The IT power does not include facility overhead such as cooling and power distribution. To estimate facility-level demand, the IT power is multiplied by the selected PUE value:

$$P_{\mathrm{facility,center}}(t) = P_{\mathrm{IT,center}}(t) \cdot \mathrm{PUE}. \tag{4.9}$$

The resulting value represents the estimated total facility power demand of one simulated HPC center. Since pandapower load values are assigned in megawatts, the facility power is converted from watts to megawatts:

$$P_{\mathrm{MW,center}}(t) = \frac{P_{\mathrm{facility,center}}(t)}{10^6}. \tag{4.10}$$

This center-level profile is the main load signal used for the grid simulation.

4.6. Multi-Center Load Modeling

To study larger scenarios, the center-level load profile is scaled to multiple simulated HPC centers. If all centers are active at the same time, the total simulated facility demand is

$$P_{\mathrm{total}}(t) = N_{\mathrm{centers}} \cdot P_{\mathrm{facility,center}}(t), \tag{4.11}$$

where N_{centers} is the number of simulated HPC centers.

For load distribution scenarios, not every center is necessarily active at every time step. This is represented using an activity variable

$$a_c(t) \in \{0, 1\}, \tag{4.12}$$

where $a_c(t) = 1$ means that center c is active at time t, and $a_c(t) = 0$ means that it is inactive. The total active facility demand can then be written as

$$P_{\text{total}}(t) = \sum_{c=1}^{N_{\text{centers}}} a_c(t) \cdot P_{\text{facility,center}}(t). \tag{4.13}$$

This representation is important for the load distribution scenario evaluated later. It allows the framework to distinguish between full simultaneous operation and scheduled operation in which only a subset of centers is active at a given time.

4.7. Workload Profile Used for Simulation

The final output of the workload modeling step is a time-dependent load profile that can be assigned to pandapower load elements. The profile contains the estimated facility-level power demand for one simulated center and, depending on the scenario, activity indicators for multiple centers.

The workload profile includes the following main quantities:

- elapsed time and sampling interval,
- measured or averaged GPU power,
- estimated node-level power,
- estimated center-level IT power,
- estimated center-level facility power,
- converted load values in megawatts,
- optional center activity indicators for scheduling scenarios.

The modeling process intentionally separates measured workload behavior from scaling assumptions. The measured GPU trace provides the dynamic part of the profile, while non-GPU node power, number of nodes, PUE, number of centers, and scheduling assumptions are configurable. This separation makes the simulation framework flexible and allows different workload and infrastructure scenarios to be compared systematically.

5. Simulation Methodology

This chapter describes how the workload profiles developed in Chapter 4 are used in the power system simulation. The focus is on the conceptual simulation methodology rather than on software implementation details. The implementation itself is described in Chapter 6.

5.1. Overview of the Simulation Approach

The simulation framework connects workload-level GPU power behavior with grid-level power flow analysis. The workload model produces time-dependent facility-level load profiles for one or more simulated HPC centers. These load profiles are then assigned to electrical loads in a grid model and evaluated using AC power flow calculations.

The simulation process consists of four main steps. First, measured GPU power traces are processed and converted into node-level power profiles. Second, these node-level profiles are scaled to HPC-center-level facility demand using assumptions about node count, non-GPU node power, and facility overhead. Third, the resulting load profiles are assigned to a synthetic grid model or a SimBench-based benchmark grid. Finally, AC power flow simulations are executed to evaluate voltage levels, transformer loading, line loading, peak demand, energy consumption, and operational cost.

This structure makes it possible to compare different workload and infrastructure scenarios under controlled assumptions. The framework does not attempt to reproduce one specific real HPC center in detail. Instead, it provides a reproducible method for studying how workload-derived power demand behaves when scaled to larger HPC-like environments and applied to electrical grid models.

5.2. Synthetic Grid Model

The first grid representation used in this thesis is a synthetic grid model. It is designed to provide a controlled and transparent environment for evaluating workload-derived HPC loads. The model uses voltage levels commonly found in German power systems, including a high-voltage connection level, a medium-voltage distribution level, and a low-voltage load level.

The synthetic grid follows a simplified connection structure. A 110 kV external grid connection supplies a 110/20 kV transformer. The medium-voltage level is represented at 20 kV and connects to lower-voltage infrastructure through 20/0.4 kV transformers. The simulated HPC loads are represented at the 0.4 kV level, where rack-level or aggregated load points are assigned.

The synthetic model is not intended to reproduce a real German grid topology or a specific HPC facility connection. Instead, it provides a representative abstraction that allows the influence of HPC-center load profiles to be studied under clearly defined assumptions. Its main advantage is controllability: the voltage levels, connection points, number of simulated centers, and load distribution can be configured directly.

5.3. SimBench-Based Grid Model

In addition to the synthetic grid model, the simulation framework uses a benchmark-based grid model derived from SimBench. SimBench provides standardized power system benchmark networks for reproducible power flow studies [6]. In this thesis, the SimBench-based grid is used as an additional reference model to evaluate the workload-derived HPC loads in a less manually constructed network environment.

The SimBench model is not used to represent a specific HPC site. Instead, it provides a standardized network structure into which simulated HPC loads can be inserted. This allows the results from the synthetic grid model to be compared with results obtained from a benchmark grid. The synthetic model offers direct control over voltage levels and load placement, while the SimBench model provides an established benchmark topology.

For the purpose of this thesis, the workload-derived load profiles are assigned to selected buses in the SimBench network. These additional loads represent simulated HPC centers and are updated according to the workload profile. The same simulation procedure is then applied as in the synthetic grid case, which allows both grid models to be evaluated using the same workload assumptions and scenario definitions.

Using both grid representations supports the methodological aim of the thesis. The synthetic grid provides a transparent baseline for controlled scenario analysis, while the SimBench-based grid helps assess whether the developed workload-to-grid simulation approach can also be applied in a standardized benchmark environment.

5.4. Assignment of HPC Loads to the Grid

The facility-level HPC load profiles calculated in Chapter 4 are assigned to electrical loads in the grid model. For each time step, the active power value of the simulated HPC center is converted to megawatts and assigned to the corresponding pandapower load element.

If the HPC-center load is distributed across multiple rack-level load points, the total center load is divided across these load points. A simplified distribution can be written as

$$P_{\text{rack}}(t) = \frac{P_{\text{facility,center}}(t)}{N_{\text{racks}}}, \tag{5.1}$$

where $P_{\text{rack}}(t)$ is the active power assigned to one rack-level load and N_{racks} is the number of rack-level load points. This approach allows the total HPC-center demand to be represented across several electrical load elements instead of only one aggregated load.

Reactive power is represented using a simplified proportional assumption. In the implementation, reactive power is derived from active power using a fixed factor:

$$Q(t) = k_Q \cdot P(t), \tag{5.2}$$

where $Q(t)$ is the reactive power, $P(t)$ is the active power, and k_Q is a constant proportional factor. This is a simplification and does not model detailed power factor behavior of real HPC equipment. However, it allows the workload-derived loads to be represented in AC power flow simulations, where both active and reactive power are required.

5.5. Time-Series Power Flow Simulation

The simulation is performed as a time-series calculation. For each time step of the workload profile, the corresponding load values are assigned to the grid model. An AC power flow calculation is then executed for that time step. The results are stored and later analyzed as time-dependent output metrics.

For each time step t_i, the simulation performs the following operations:

1. Update the active and reactive power values of the HPC load elements.
2. Execute an AC power flow calculation using pandapower.
3. Extract bus voltages, transformer loading, line loading, and convergence status.
4. Store the results for later evaluation.

This procedure preserves the temporal behavior of the workload-derived load profile. Short peaks and changes in workload demand can therefore influence the calculated grid state instead of being hidden by a single average power value.

The detailed simulation mode uses AC power flow because voltage behavior and reactive power effects are relevant for the evaluation. A faster approximation mode can be used for exploratory runs, but the main evaluation focuses on AC power flow results.

5.6. Evaluation Metrics

The simulation framework evaluates both workload-related and grid-related metrics. The workload-related metrics describe power demand, accumulated energy, and operational cost. The grid-related metrics describe the electrical state of the simulated network.

The most important metrics are summarized in Table 5.1.

Table 5.1.: Main evaluation metrics used in the simulation framework.

Metric	**Purpose**
Total energy consumption	Measures accumulated workload or facility energy demand
Average power	Describes mean power demand over the evaluated trace
Peak power	Captures maximum instantaneous demand
Operational cost	Estimates electricity cost under time-dependent prices
Minimum bus voltage	Evaluates voltage behavior in the grid model
Maximum transformer loading	Measures transformer utilization relative to rated capacity
Maximum line loading	Measures line utilization relative to rated capacity
Power flow convergence	Indicates numerical feasibility of the simulated grid state
Violation counts	Reports whether voltage, line, or transformer limits are exceeded

Energy is calculated by integrating the simulated power over the sampling intervals of the workload trace. Grid-related metrics are extracted from the pandapower result tables after each power flow calculation. The combination of these metrics allows the evaluation to consider both energy behavior and electrical network behavior.

5.7. Operational Scenarios

Several operational scenarios are considered in order to evaluate how different assumptions influence energy demand, peak power, operational cost, and grid-related behavior. The scenarios are summarized in Table 5.2.

Table 5.2.: Overview of the operational scenarios considered in the simulation.

Scenario	**Main change**	**Expected effect**	**Interpretation**
Baseline	Original workload profile	Reference case	No optimization applied
GPU power limit	Lower GPU power component	Lower peak power	Runtime may increase
PUE efficiency	Lower facility overhead	Lower facility power	Runtime unchanged
Workload shifting	Shift execution time	Cost or overlap changes	Energy mostly unchanged
Load distribution	Limit simultaneous active centers	Lower concurrent peak	Same-work energy must be checked

The baseline scenario uses the original workload-derived load profile without additional operational changes. It serves as the reference case against which all other scenarios are compared.

The GPU power limiting scenario reduces the GPU power component by applying a power factor. This can reduce instantaneous power demand and peak load. However, if the reduced power limit causes the workload to run longer, the effect on total energy consumption may be smaller than the reduction in power alone suggests.

The PUE-based facility efficiency scenario changes the facility overhead by adjusting the PUE-related part of the facility power. A lower overhead reduces the total facility power required for the same IT load, but it does not directly change the computational runtime of the workload.

The workload shifting scenario changes the execution time of the workload profile. This does not necessarily reduce total energy consumption, but it can affect operational cost when time-dependent electricity prices are used. It can also change the overlap between several simulated centers.

The load distribution scenario changes which simulated HPC centers are active at each time step. In the implemented rotating strategy, a center is either active and receives the full center-level workload demand, or inactive and receives no workload demand. This can reduce simultaneous peak demand, but it must be interpreted on a same-work basis because reducing activity within a fixed time window does not automatically reduce the total energy required to complete the same amount of work.

5.8. Capacity Analysis

In addition to the operational scenarios, the framework includes a capacity analysis mode. This mode is not an optimization scenario. Instead, it evaluates how the selected grid model behaves when the number of simulated HPC centers is increased.

The capacity analysis uses the peak-load time step of the workload profile. For each tested number of centers, the peak load is applied to the grid model and an AC power flow calculation is executed. A tested configuration is considered safe if the power flow converges and the selected voltage and loading limits are not violated.

The capacity analysis checks the following conditions:

- minimum bus voltage remains above the selected lower voltage limit,
- line loading remains below the selected maximum loading limit,
- transformer loading remains below the selected maximum loading limit,
- the AC power flow calculation converges.

If no failure occurs within the tested range, the result is interpreted as the largest safe tested value, not as an absolute physical capacity limit. This distinction is important because the result depends on the chosen grid model, workload assumptions, tested center range, and technical limits.

5.9. Cost Modeling

In addition to technical metrics such as energy consumption and peak power demand, the simulation framework estimates operational electricity cost. This is relevant because two workload executions with the same total energy consumption can lead to different costs if they are executed at different times and electricity prices vary over the day.

The cost model combines the time-dependent workload power profile with an electricity price profile. For each time step, the consumed energy is calculated from the power value and the sampling interval. This energy is then multiplied by the electricity price assigned to that time step. The total cost is calculated as

$$C = \sum_{i=1}^{n} E(t_i) \cdot p_{\mathrm{el}}(t_i), \tag{5.3}$$

where C denotes the total electricity cost, $E(t_i)$ is the energy consumed during time step t_i, and $p_{\mathrm{el}}(t_i)$ is the electricity price at that time step.

The energy for each time step is derived from the simulated power demand:

$$E(t_i) = P(t_i) \cdot \Delta t_{h,i}, \tag{5.4}$$

where $P(t_i)$ is the power demand at time step t_i, and $\Delta t_{h,i}$ is the sampling interval in hours. If $P(t_i)$ is given in kilowatts and $p_{\mathrm{el}}(t_i)$ is given in euros per kilowatt-hour, the resulting cost is obtained in euros.

The cost model is not intended to represent a complete electricity market or billing model. It does not include grid fees, taxes, demand charges, balancing costs, or contractual procurement structures. Instead, it provides a simplified operational cost estimate that allows workload timing scenarios to be compared.

5.10. Methodological Scope

The simulation methodology is based on representative modeling assumptions. The available workload traces describe single-GPU executions and are scaled to larger HPC-like environments. The synthetic grid uses representative voltage levels rather than a site-specific topology, while the SimBench-based grid provides a standardized benchmark model. Facility overhead is modeled through PUE, and reactive power is approximated using a simplified proportional assumption.

These simplifications are necessary because detailed real-world HPC infrastructure data was only partially available. The objective of the methodology is therefore not to reconstruct one real facility exactly, but to provide a reproducible scenario-based simulation framework. Within this scope, the framework can be used to analyze how workload-derived power profiles influence energy demand, peak load, operational cost, voltage behavior, transformer loading, and line loading under controlled assumptions.

The results of the simulation should therefore be interpreted as scenario-based estimates rather than direct measurements of an existing HPC center. This distinction is important for the interpretation of the evaluation chapter. The framework is intended to support comparative analysis between scenarios, not to make site-specific operational claims about a particular real-world facility.

6. Implementation

This chapter describes the implementation of the simulation framework. While Chapter 5 introduced the methodological concepts, this chapter explains how these concepts are realized in the Python-based simulation pipeline. The implementation is organized into modular components for trace processing, power modeling, grid construction, time-series simulation, scenario handling, capacity analysis, cost estimation, and dashboard visualization.

6.1. Simulation Structure

The implementation consists of a Streamlit dashboard and a Python simulation package. The dashboard provides the user interface for selecting data, parameters, grid backends, and operational scenarios. The simulation package contains the core calculation logic. This separation allows the modeling and simulation functions to be used independently from the dashboard interface.

The main components of the simulation framework are summarized in Table 6.1. For readability, the table lists only the file names. Most implementation files are located in the `simulation/` package, while the dashboard is implemented in `app.py`.

Table 6.1.: Main components of the simulation framework.

File	Purpose
app.py	Streamlit dashboard for parameter selection, simulation execution, and visualization.
profile _builder.py	Loads, validates, aligns, and averages CSV-based GPU power traces.
power_model.py	Converts GPU power traces into node-level, center-level, and facility-level power demand.
grid_model.py	Builds the synthetic grid and integrates the SimBench-based grid model.
run _simulation.py	Executes pandapower simulations and extracts grid-related metrics.
optimization _scenarios.py	Implements GPU power limiting, workload shifting, PUE changes, and load distribution.
cost_model.py	Estimates operational cost using time-dependent electricity prices.
capacity _analysis.py	Performs hosting capacity analysis for increasing simulated center counts.
energy _projection.py	Calculates energy and cost projections for selected time windows.
raw_runs/	Contains CSV traces for training and inference workloads.

The modular structure separates data processing, workload modeling, grid construction, simulation execution, scenario handling, cost estimation, and visualization. This is important for reproducibility because each step can be inspected independently. It also makes the framework extendable: for example, newer workload traces can be added without changing the grid simulation logic, and additional grid models can be integrated without changing the trace preprocessing step.

6.2. Input Trace Processing

Input trace processing is implemented in `profile_builder.py`. This module loads CSV files from the selected workload directory and converts them into a consistent time-series profile. The available workload data consists of training and inference traces stored as CSV files.

The module checks whether the required columns are available. The most important input column is the GPU power measurement, which is stored in watts and represents the time-dependent power behavior of a single-GPU workload execution. Optional columns such as GPU utilization, CPU utilization, number of GPUs, epoch, and step are used when available, but the simulation mainly relies on the timestamp and power columns.

During preprocessing, timestamps are converted into a time format and sorted in chronological order. The implementation then calculates the time difference between consecutive samples and stores this value as `delta_seconds`. This value is later used for energy calculation. If invalid or missing intervals occur, a fallback value based on the median sampling interval is used. This prevents individual timestamp irregularities from breaking the simulation pipeline.

When multiple CSV runs are available, the traces are aligned and converted into a representative workload profile. The implementation trims the runs to a common length and averages the relevant power values across the available runs. In the baseline evaluation, this produces a representative training profile from four measured runs.

The implementation supports training, inference, and combined workload modes. In training mode, the representative profile is derived from training traces. In inference mode, it is derived from inference traces. In combined mode, an inference pattern can be added to the training profile. This combined mode is an assumption-based scenario and is not interpreted as a direct measurement of a real combined deployment.

6.3. Power Model Implementation

The conversion from workload-level GPU power to facility-level power is implemented in `power_model.py`. The input to this module is the processed workload profile produced by the trace processing step.

For each time step, the implementation calculates node-level power by adding assumed non-GPU components to the measured GPU power. These components include CPU power, memory power, storage power, and network power. The GPU trace therefore provides the dynamic part of the node power, while the remaining components are represented by configurable assumptions.

The node-level power is then multiplied by the number of nodes per simulated HPC center. This produces the modeled IT power demand of one center. Facility overhead is included by multiplying the IT power with the selected PUE value. The implementation also stores the overhead power separately as the difference between facility power and IT power.

Finally, the facility-level power is converted from watts to megawatts, because pandapower represents active load power in megawatts. The resulting column, `center_total_power_mw`, is used by the grid simulation and capacity analysis modules.

This implementation separates measurement data from modeling assumptions. The GPU power trace remains the measured time-dependent signal, while node count, non-GPU node power, PUE, and center count are configurable parameters. This allows the same trace to be evaluated under different infrastructure assumptions.

6.4. Grid Model Implementation

The grid model implementation is contained in `grid_model.py`. This module provides the electrical network models used for the simulations. The framework supports two grid backends: a synthetic HPC-oriented grid model and a SimBench-based benchmark grid.

The synthetic grid model is constructed as an assumption-based representation of an HPC connection structure. It includes an external high-voltage grid connection, transformation to a medium-voltage level, a campus distribution level, lower-voltage transformers, and rack-level load points. The model is not intended to reproduce a specific real facility. Its purpose is to provide a transparent grid structure in which connection points and load scaling can be controlled directly.

The SimBench-based backend loads a standardized benchmark network and inserts simulated HPC loads at selected connection points. This backend is used to evaluate the same workload-derived load profiles in a benchmark-oriented grid environment. As with the synthetic grid, the SimBench-based model is not interpreted as a real HPC site, but as a reproducible reference grid.

Both grid backends return a pandapower network object and a mapping between simulated HPC centers and pandapower load elements. This mapping is required by the simulation loop because the active and reactive power values of these load elements are updated at each time step.

6.5. Time-Series Simulation Loop

The time-series simulation is implemented in `run_simulation.py`. For each row of the workload profile, the simulation reads the center-level facility power and assigns it to the corresponding pandapower load elements. If multiple rack-level load elements are used, the center power is divided across these loads.

For each load element, active power is assigned in megawatts. Reactive power is estimated using a simplified proportional assumption, where reactive power is calculated as a fixed fraction of active power. This keeps the reactive power model transparent while allowing the loads to be used in AC power flow calculations.

After updating the load values, the implementation executes a power flow calculation. In the detailed mode, an AC power flow is performed using `pp.runpp`. The framework also supports a faster exploratory mode using a DC approximation. The AC mode is used for the main evaluation because voltage behavior and reactive power effects are relevant for the grid-related metrics.

For each time step, the implementation stores the main simulation outputs. These include total active power, number of active centers, minimum and maximum voltage, maximum transformer loading, maximum line loading, grid losses, external grid supply, convergence status, and violation counts. The result is returned as a time-series dataframe for later analysis and visualization.

6.6. Energy and Grid Metric Calculation

Energy is calculated by multiplying simulated power with the duration of each time step. Since the time step duration is stored as `delta_seconds`, it is first converted to hours. The energy for each step is then accumulated over the trace.

Grid metrics are extracted from the pandapower result tables after each power flow calculation. Bus voltage magnitudes are taken from `net.res_bus`, transformer loading from `net.res_trafo`, and line loading from `net.res_line`. The implementation also records whether the power flow converged successfully.

In addition to the main metrics, diagnostic values such as active losses, reactive power, external grid supply, and violation counts are stored. These values are useful for interpreting whether a scenario produces numerical or operational stress in the selected grid model.

6.7. Scenario Implementation

Operational scenarios are implemented mainly in `optimization_scenarios.py`. The scenario functions modify the workload profile, scaling assumptions, or activity pattern before the simulation is executed.

The GPU power limiting scenario reduces the GPU power component by applying a power factor. If runtime slowdown is enabled, the implementation extends the trace duration according to the selected slowdown model. This reflects the idea that lower GPU power may reduce instantaneous demand while increasing execution time.

The PUE-based facility efficiency scenario changes the facility overhead. In the implemented model, the cooling overhead factor is applied to the non-IT overhead part of the facility power. This affects facility-level energy and cost, but it does not directly alter workload runtime.

The workload shifting scenario modifies the time placement of the workload profile. This is especially relevant for the cost model, because the same energy demand can lead to different costs depending on the electricity price assigned to each time step.

The load distribution scenario changes which simulated centers are active at each time step. For each time step, a center can be active and receive the center-level load, or inactive and receive zero workload demand. This makes it possible to evaluate how limiting simultaneous activity affects peak load and grid-related metrics.

6.8. Capacity Analysis Implementation

Capacity analysis is implemented in `capacity_analysis.py`. Unlike the operational scenarios, this mode does not optimize or modify the workload. Instead, it tests how the selected grid model behaves when the number of simulated HPC centers is increased.

The implementation uses the peak-load time step of the workload profile. For each tested center count, a grid model is created, the peak workload is assigned to the simulated HPC loads, and an AC power flow calculation is executed. The result is classified as safe if the power flow converges and the selected voltage and loading limits are not violated.

The capacity analysis records the tested number of centers, applied peak power, minimum voltage, maximum line loading, maximum transformer loading, convergence status, and possible failure reason. If no failure occurs within the tested range, the reported result is interpreted as the largest safe tested value, not as an absolute physical capacity limit.

6.9. Cost Model Implementation

The cost model is implemented in `cost_model.py`. It combines the time-dependent simulated power demand with an electricity price profile. For each time step, the consumed energy is calculated from the power value and the sampling interval. This energy is then multiplied by the electricity price assigned to the corresponding time period.

The cost model supports the comparison of baseline and shifted workload scenarios. It is not designed as a complete electricity market or billing model. It does not include grid fees, taxes, demand charges, or contractual pricing structures. Instead, it provides an operational cost estimate that is sufficient for comparing workload timing scenarios under simplified time-dependent price assumptions.

6.10. Dashboard Implementation

The interactive dashboard is implemented in `app.py` using Streamlit. It provides a user interface for selecting workload data, scaling parameters, grid backend, scenario settings, capacity analysis options, and visualization outputs. The dashboard connects the individual simulation modules into a complete workflow.

The dashboard allows the user to configure parameters such as workload mode, number of simulated HPC centers, nodes per center, clusters per center, racks per cluster, PUE, CPU power per node, grid backend, and simulation mode. It then executes the workload modeling and grid simulation pipeline and displays the resulting metrics and plots.

The dashboard is mainly used as an analysis and presentation tool. It helps inspect how parameter changes affect energy consumption, peak power, voltage behavior, transformer loading, line loading, capacity limits, and operational cost. The underlying calculations are performed by the modular simulation functions described in the previous sections.

6.11. Implementation Scope

The implementation is designed as a reproducible research prototype. It is suitable for scenario-based analysis and for exploring the relationship between ML workload traces and grid-level simulation results. It is not intended to replace detailed electrical planning software or measured site-specific validation.

Several implementation choices are simplified by design. Non-GPU node power is represented through constant assumptions, PUE is used as an aggregated facility multiplier, and reactive power is estimated using a proportional factor. These choices keep the simulation transparent and configurable while still allowing the main research question to be investigated.

The implementation therefore supports the objective of this thesis: to build a flexible simulation framework that links workload-derived power demand with power flow analysis, energy evaluation, capacity analysis, and operational cost estimation under controlled assumptions.

7. Results and Evaluation

This chapter presents the results obtained from the simulation framework. The evaluation focuses on workload-derived power demand, energy consumption, peak power, voltage behavior, transformer loading, line loading, operational cost, capacity behavior, and the quality of the input traces. The purpose of the chapter is to evaluate how measured single-GPU workload traces behave after being scaled to simulated HPC-center level and applied to the selected grid models.

The results are interpreted as scenario-based simulation outputs. They are not direct measurements of a real HPC center or a real German power grid. Instead, they show how workload-derived load profiles affect the synthetic and benchmark-based grid models under the assumptions described in the previous chapters.

7.1. Evaluation Setup and Input Trace Quality

The first evaluation run is used as the baseline scenario. It uses the measured training traces from the available CSV files and applies the workload model without any additional optimization or scheduling strategy. The baseline therefore provides the reference case for the later comparison with GPU power limiting, PUE-based facility efficiency changes, workload shifting, load distribution, capacity analysis, and the SimBench-based grid model.

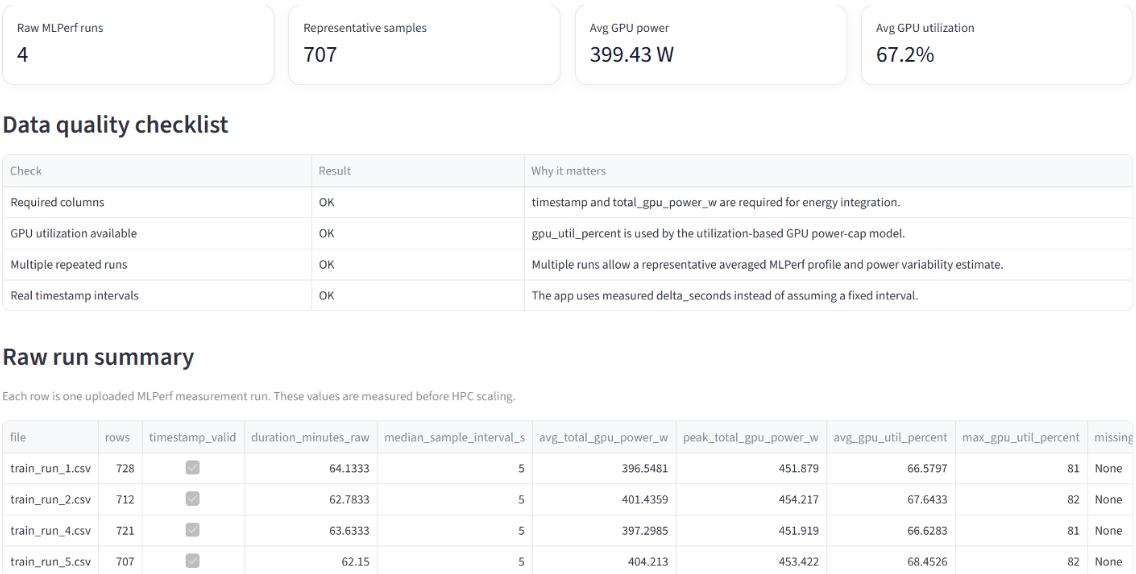
For the baseline run, the workload profile is derived from the training traces stored in the raw training data directory. The simulation uses 20 simulated HPC centers with 500 nodes per center, resulting in a total of 10,000 configured nodes. Facility overhead is represented through the selected PUE value, and the resulting facility-level load profile is evaluated using the synthetic grid backend with AC power flow.

Table 7.1 summarizes the main configuration of the baseline run.

Table 7.1.: Baseline evaluation setup.

Parameter	**Baseline setting**
Workload input	Training CSV traces
Raw workload runs	4 measured runs
Representative samples	707
Average GPU power	399.43 W
Average GPU utilization	67.2%
Simulated HPC centers	20
Nodes per center	500
Total configured nodes	10,000
Grid backend	Synthetic HPC grid
Simulation mode	AC power flow
Optimization or scheduling	Disabled

Before evaluating the scaled HPC load, the input traces are inspected to verify that the workload profile is based on usable measurement data. Figure 7.1 shows the measured trace pipeline from the dashboard. The selected training data consists of four raw workload runs and is converted into a representative profile with 707 samples.


Raw MLPerf runs	Representative samples	Avg GPU power	Avg GPU utilization
4	707	399.43 W	67.2%

Data quality checklist

Check	Result	Why it matters
Required columns	OK	timestamp and total_gpu_power_w are required for energy integration.
GPU utilization available	OK	gpu_util_percent is used by the utilization-based GPU power-cap model.
Multiple repeated runs	OK	Multiple runs allow a representative averaged MLPerf profile and power variability estimate.
Real timestamp intervals	OK	The app uses measured delta_seconds instead of assuming a fixed interval.

Raw run summary

Each row is one uploaded MLPerf measurement run. These values are measured before HPC scaling.

file	rows	timestamp_valid	duration_minutes_raw	median_sample_interval_s	avg_total_gpu_power_w	peak_total_gpu_power_w	avg_gpu_util_percent	max_gpu_util_percent	missing
train_run_1.csv	728	✓	64.1333	5	396.5481	451.879	66.5797	81	None
train_run_2.csv	712	✓	62.7833	5	401.4359	454.217	67.6433	82	None
train_run_4.csv	721	✓	63.6333	5	397.2985	451.919	66.6283	81	None
train_run_5.csv	707	✓	62.15	5	404.213	453.422	68.4526	82	None

Figure 7.1.: Measured workload trace pipeline used for the baseline simulation.

The data quality checks indicate that the required columns are available, GPU utilization is present, multiple repeated runs are available, and real timestamp intervals are used. This is important because the simulation does not assume a fixed artificial time step. Instead, the timestamp intervals from the CSV traces are used for the energy integration. The average GPU power of the representative trace is 399.43 W, with an average GPU utilization of 67.2%. These values describe the workload before it is scaled to node, center, and facility level.

The raw run summary also shows that the individual training traces have similar durations and median sampling intervals. The median sampling interval is 5 s for the displayed runs. This supports the use of an averaged representative trace for the baseline evaluation.

7.2. Baseline Results on the Synthetic Grid

Figure 7.2 shows the main baseline results after the single-GPU workload trace has been scaled to the simulated HPC-center configuration. Since no optimization or scheduling strategy is active, the current scenario values are identical to the baseline values.

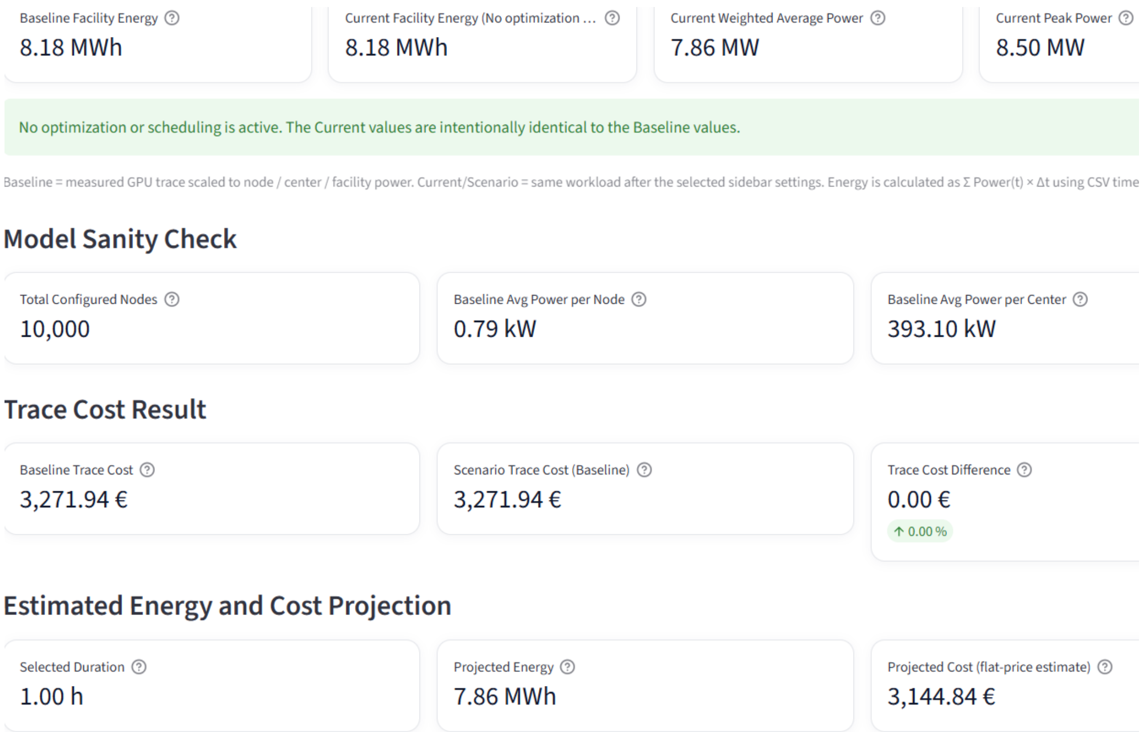


Figure 7.2.: Baseline overview for the synthetic-grid simulation without optimization or scheduling.

The baseline facility energy over the measured trace is 8.18 MWh. The weighted average power is 7.86 MW, while the peak power reaches 8.50 MW. The dashboard also reports 10,000 configured nodes, which is consistent with the selected setup of 20 simulated HPC centers and 500 nodes per center. The average power per node is 0.79 kW, and the average power per center is 393.10 kW.

These values are internally consistent. Dividing the average facility power of 7.86 MW by 10,000 nodes gives approximately 0.786 kW per node, which corresponds to the reported average power per node. Similarly, dividing 7.86 MW by 20 simulated centers gives approximately 393 kW per center. This indicates that the scaling from node level to center level and multi-center level is applied consistently in the baseline run.

The baseline trace cost is 3,271.94 EUR under the selected electricity price assumptions. Since no optimization or scheduling is active, the scenario trace cost is identical to the baseline cost, and the cost difference is 0.00 EUR. The projected energy for a selected duration of one hour is 7.86 MWh, which corresponds to the reported weighted average power. The measured trace energy of 8.18 MWh is slightly higher because it is calculated over the actual trace duration rather than an exactly one-hour projection window.

Table 7.2 summarizes the main baseline facility-level results.

Table 7.2.: Baseline facility-level simulation results.

Metric	Baseline value
Baseline facility energy	8.18 MWh
Weighted average power	7.86 MW
Peak power	8.50 MW
Total configured nodes	10,000
Average power per node	0.79 kW
Average power per center	393.10 kW
Baseline trace cost	3,271.94 EUR
Projected energy for 1 hour	7.86 MWh
Projected cost for 1 hour	3,144.84 EUR

The baseline load profile is applied to the synthetic pandapower grid and evaluated at the peak-load snapshot. Figure 7.3 shows the grid health indicators reported by the dashboard.

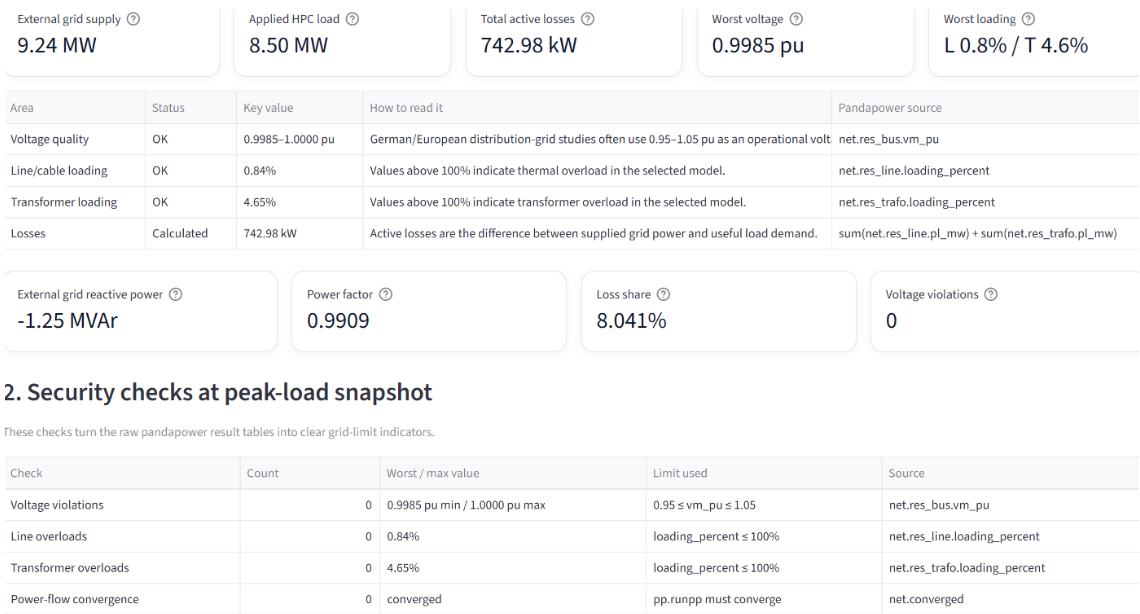


Figure 7.3.: Grid health indicators at the baseline peak-load snapshot.

At the peak-load snapshot, the applied HPC load is 8.50 MW. The external grid supplies

9.24 MW, and the total active losses are 742.98 kW. The worst voltage is 0.9985 pu, which remains close to nominal voltage. The maximum line loading is 0.8%, and the maximum transformer loading is 4.6%. No voltage violations are reported.

The security checks also indicate that no line overloads or transformer overloads occur in the baseline run. The power flow converges successfully. This means that, under the selected baseline assumptions, the synthetic grid model is not stressed by the applied HPC load. The low component loading values show that the selected synthetic grid has substantial remaining capacity for this scenario.

Table 7.3 summarizes the main grid-related results.

Table 7.3.: Baseline grid-related results at the peak-load snapshot.

Metric	**Baseline value**
Applied HPC load	8.50 MW
External grid supply	9.24 MW
Total active losses	742.98 kW
Worst voltage	0.9985 pu
Maximum line loading	0.8%
Maximum transformer loading	4.6%
External grid reactive power	-1.25 MVAr
Power factor	0.9909
Loss share	8.041%
Voltage violations	0
Line overloads	0
Transformer overloads	0
Power flow convergence	Converged

The reported loss share should be interpreted within the context of the synthetic grid model. It is useful for comparing scenarios within the same model, but it should not be interpreted as a measured loss value of a real German HPC connection. The main conclusion from the baseline grid-health results is that the modeled system remains within the selected voltage and loading limits for the baseline load level.

Figure 7.4 shows the synthetic pandapower topology used in the baseline evaluation. The topology contains the external grid, grid buses, transformer markers, HPC center connection buses, and line or cable connections. The orange nodes represent HPC center connection buses, while the external grid is shown at the center of the topology.

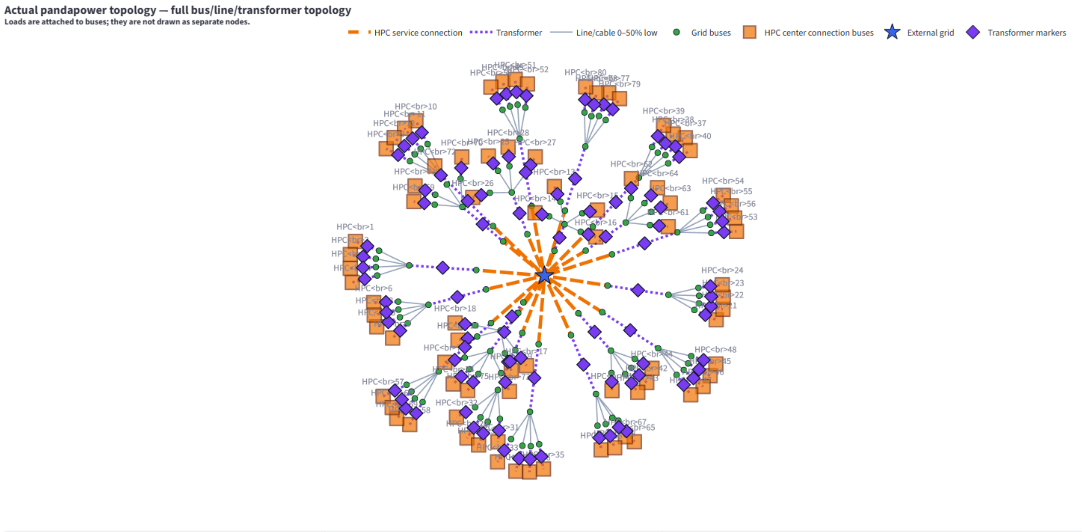


Figure 7.4.: Synthetic pandapower grid topology with simulated HPC connection buses.

The topology visualization is useful for understanding how the workload-derived loads are connected to the synthetic grid. Loads are attached to buses and are therefore not drawn as separate nodes. The figure shows that the synthetic grid is a controlled modeling environment rather than a site-specific real grid.

7.3. Operational Scenario Results

This section compares the operational scenarios against the baseline run. The scenarios are evaluated using the same workload and synthetic grid assumptions unless stated otherwise.

7.3.1. GPU Power Limiting

The GPU power limiting scenario evaluates how reducing the GPU power component affects facility-level energy demand, peak power, operational cost, and grid-related behavior. In this run, the GPU power factor is set to 0.90, meaning that the GPU-related power component is reduced by 10%. The utilization-based runtime model is enabled, so the workload duration can increase when the effective GPU performance is reduced.

Figure 7.5 shows the dashboard overview for the GPU power limiting scenario. Compared with the baseline, the scenario facility energy decreases from 8.18 MWh to 8.02 MWh. The weighted average power decreases from 7.86 MW to 7.35 MW, and the peak power decreases from 8.50 MW to 7.92 MW. The trace cost decreases from 3,271.94 EUR to 3,209.37 EUR.

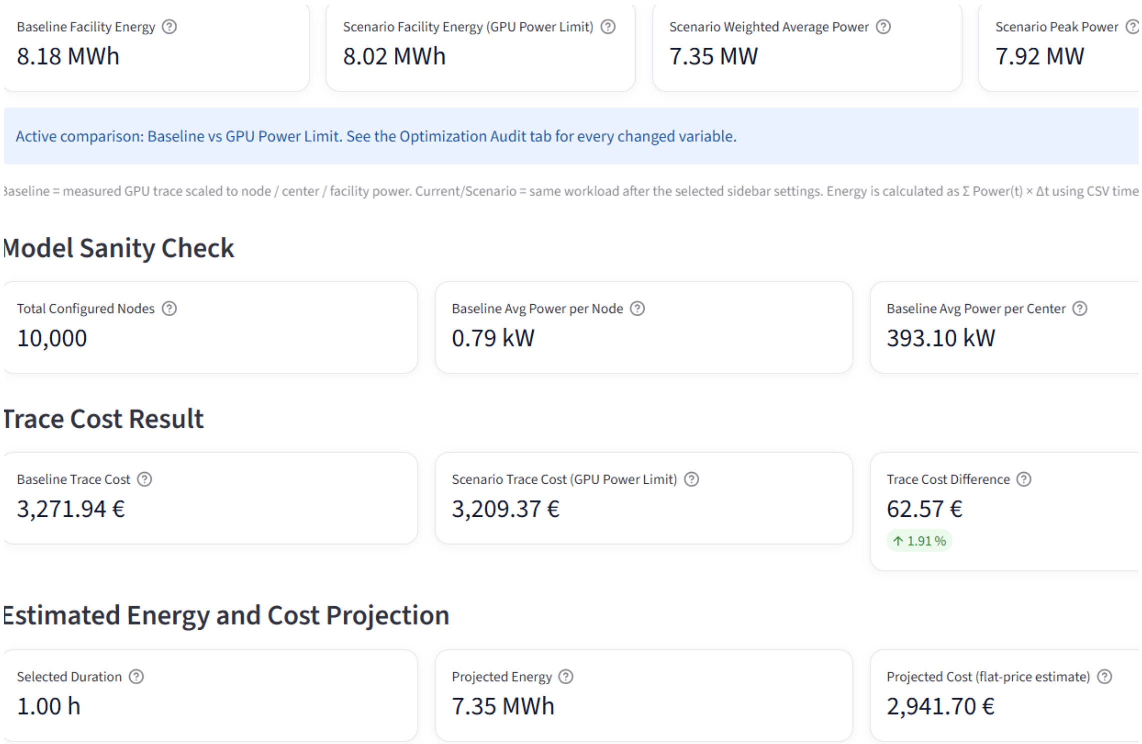
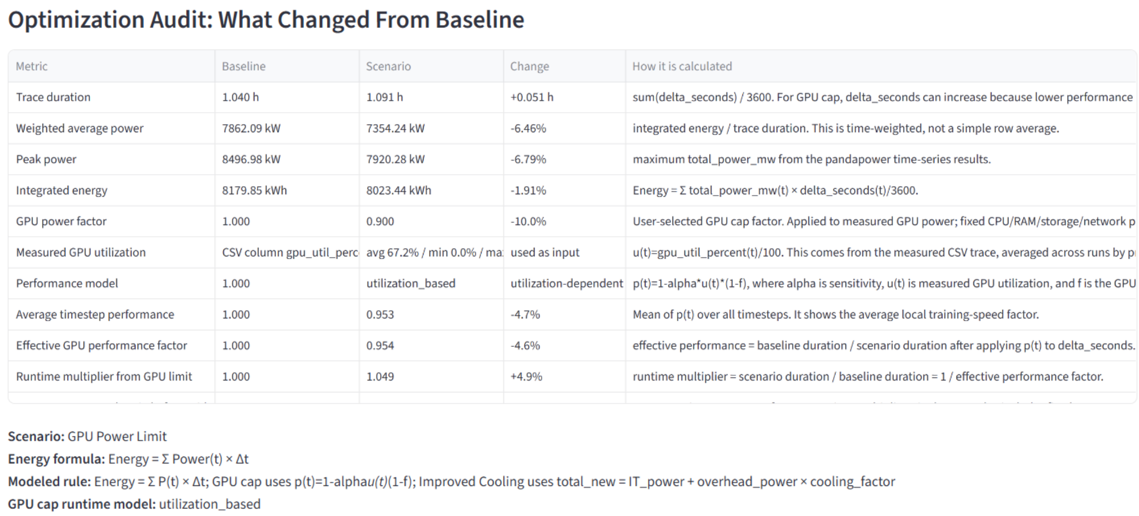


Figure 7.5.: Overview of the GPU power limiting scenario.

The optimization audit in Figure 7.6 explains why the reduction in energy is smaller than the reduction in GPU power. Although the GPU power factor is reduced by 10%, the trace duration increases from 1.040 h to 1.091 h. This corresponds to a runtime multiplier of 1.049, or an increase of approximately 4.9%. As a result, the integrated energy decreases by only 1.91%, while the weighted average power decreases by 6.46% and the peak power decreases by 6.79%.



Optimization Audit: What Changed From Baseline

Metric	Baseline	Scenario	Change	How it is calculated
Trace duration	1.040 h	1.091 h	+0.051 h	sum(delta_seconds) / 3600. For GPU cap, delta_seconds can increase because lower performance
Weighted average power	7862.09 kW	7354.24 kW	-6.46%	integrated energy / trace duration. This is time-weighted, not a simple row average.
Peak power	8496.98 kW	7920.28 kW	-6.79%	maximum total_power_mw from the pandapower time-series results.
Integrated energy	8179.85 kWh	8023.44 kWh	-1.91%	Energy = Σ total_power_mw(t) × delta_seconds(t)/3600.
GPU power factor	1.000	0.900	-10.0%	User-selected GPU cap factor. Applied to measured GPU power; fixed CPU/RAM/storage/network p
Measured GPU utilization	CSV column gpu_util_perc	avg 67.2% / min 0.0% / ma	used as input	u(t)=gpu_util_percent(t)/100. This comes from the measured CSV trace, averaged across runs by p
Performance model	1.000	utilization_based	utilization-dependent	p(t)=1-alpha*u(t)*(1-f), where alpha is sensitivity, u(t) is measured GPU utilization, and f is the GPU
Average timestep performance	1.000	0.953	-4.7%	Mean of p(t) over all timesteps. It shows the average local training-speed factor.
Effective GPU performance factor	1.000	0.954	-4.6%	effective performance = baseline duration / scenario duration after applying p(t) to delta_seconds.
Runtime multiplier from GPU limit	1.000	1.049	+4.9%	runtime multiplier = scenario duration / baseline duration = 1 / effective performance factor.

Scenario: GPU Power Limit
Energy formula: Energy = Σ Power(t) × Δt
Modeled rule: Energy = Σ P(t) × Δt; GPU cap uses p(t)=1-alphau(t)(1-f); Improved Cooling uses total_new = IT_power + overhead_power × cooling_factor
GPU cap runtime model: utilization_based

Figure 7.6.: Optimization audit for the GPU power limiting scenario.

Figure 7.7 shows the grid-health indicators for the GPU power limiting scenario. At the peak-load snapshot, the applied HPC load is 7.92 MW, compared with 8.50 MW in the baseline case. The external grid supply decreases to 8.66 MW, and the worst voltage remains close to nominal at 0.9986 pu. The maximum line loading is 0.78%, and the maximum transformer loading is 4.34%. No voltage violations, line overloads, or transformer overloads are reported.

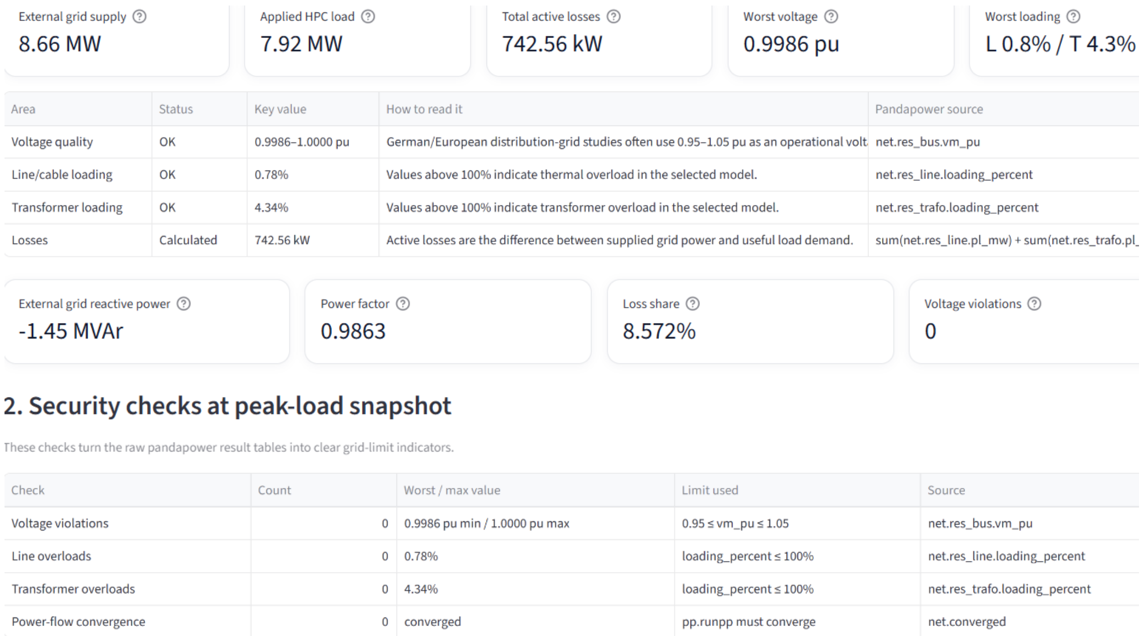


Figure 7.7.: Grid-health indicators for the GPU power limiting scenario.

Table 7.4 summarizes the comparison between the baseline and the GPU power limiting scenario.

Table 7.4.: Comparison between baseline and GPU power limiting scenario.

Metric	Baseline	GPU limit	Change
Trace duration	1.040 h	1.091 h	+4.9%
Weighted average power	7.86 MW	7.35 MW	-6.46%
Peak power	8.50 MW	7.92 MW	-6.79%
Integrated energy	8.18 MWh	8.02 MWh	-1.91%
Trace cost	3,271.94 EUR	3,209.37 EUR	-62.57 EUR
Worst voltage	0.9985 pu	0.9986 pu	Slight improvement
Maximum line loading	0.84%	0.78%	Lower
Maximum transformer loading	4.65%	4.34%	Lower
Voltage violations	0	0	No change

Overall, the GPU power limiting scenario reduces peak demand and slightly improves the grid-related indicators in the synthetic grid model. However, because the modeled runtime increases, the energy reduction is smaller than the reduction in instantaneous GPU power. This confirms that power limiting and runtime effects must be evaluated together when estimating the energy and cost impact of ML workload operation.

7.3.2. PUE-Based Facility Efficiency

The PUE-based facility efficiency scenario evaluates the effect of reducing facility overhead while keeping the ML workload itself unchanged. In this run, the cooling overhead factor is set to 0.85. This factor is applied only to the non-IT overhead component of the facility power, not to the complete facility load. As a result, the IT power remains unchanged, while the effective PUE decreases from 1.300 to 1.255.

Figure 7.8 shows the dashboard overview for the improved cooling scenario. Compared with the baseline, the scenario facility energy decreases from 8.18 MWh to 7.90 MWh. The weighted average power decreases from 7.86 MW to 7.59 MW, and the peak power decreases from 8.50 MW to 8.20 MW. The trace cost decreases from 3,271.94 EUR to 3,158.68 EUR.

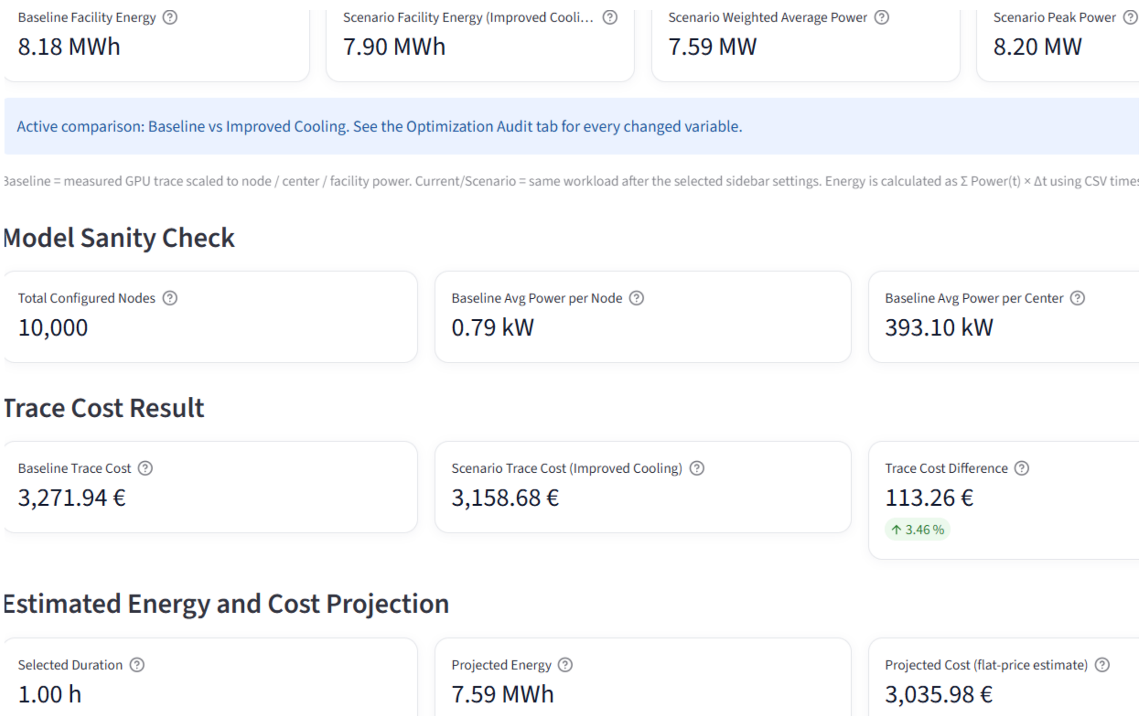


Figure 7.8.: Overview of the PUE-based facility efficiency scenario.

The important difference compared with GPU power limiting is that the workload duration remains unchanged. The trace duration stays at 1.040 h in both the baseline and the improved cooling scenario. This is expected because the scenario does not change the GPU power trace or the computational performance of the workload. Instead, it only reduces the facility overhead.

The audit results show that the cooling overhead factor reduces the non-IT overhead by

15%. Since this factor is applied only to the overhead part of the facility power, the total facility power decreases by 3.46%, not by 15%. This distinction is important because the IT load remains the dominant part of the total facility demand.

Figure 7.9 shows the grid-health indicators for the improved cooling scenario. At the peak-load snapshot, the applied HPC load is 8.20 MW. The external grid supply is 8.95 MW, and the total active losses are 742.76 kW. The worst voltage is 0.9985 pu, the maximum line loading is 0.81%, and the maximum transformer loading is 4.49%. No voltage violations are reported.

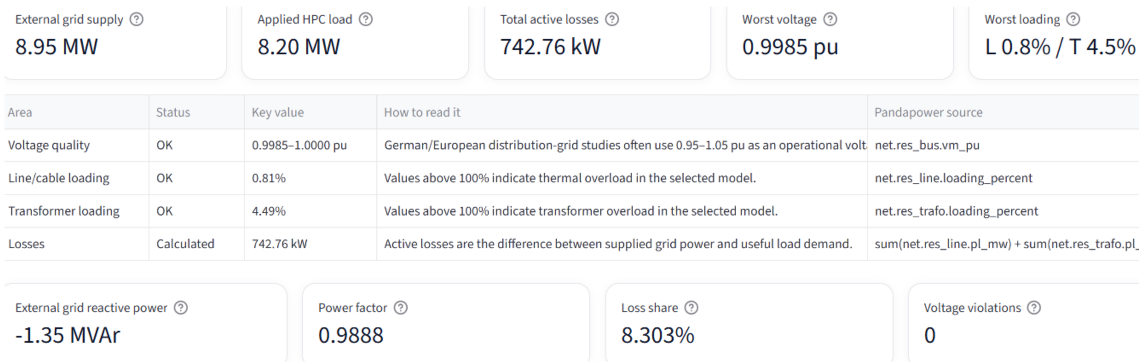


Figure 7.9.: Grid-health indicators for the PUE-based facility efficiency scenario.

Table 7.5 summarizes the comparison between the baseline and the PUE-based facility efficiency scenario.

Table 7.5.: Comparison between baseline and PUE-based facility efficiency scenario.

Metric	**Baseline**	**Improved cooling**	**Change**
Trace duration	1.040 h	1.040 h	No change
Weighted average power	7.86 MW	7.59 MW	-3.46%
Peak power	8.50 MW	8.20 MW	-3.46%
Integrated energy	8.18 MWh	7.90 MWh	-3.46%
Trace cost	3,271.94 EUR	3,158.68 EUR	-113.26 EUR
Cooling overhead factor	1.000	0.850	-15.0% overhead
Effective PUE	1.300	1.255	-3.46% facility power
Worst voltage	0.9985 pu	0.9985 pu	No relevant change
Maximum line loading	0.84%	0.81%	Lower
Maximum transformer loading	4.65%	4.49%	Lower
Voltage violations	0	0	No change

Overall, the PUE-based facility efficiency scenario reduces energy consumption, peak demand, and operational cost without increasing workload runtime. This differs from the GPU power limiting scenario, where a reduction in GPU power also introduces a runtime extension.

7.3.3. Cost Evaluation and Workload Shifting

The workload shifting scenario evaluates how the execution time of a workload affects operational cost when time-dependent electricity prices are used. In this run, the same workload profile and the same simulated HPC configuration are used as in the baseline. The difference is only the placement of the workload on the 24-hour tariff clock.

The baseline workload is placed at 18:00 and ends at 19:02. The shifted scenario places the same measured workload at 00:00, where it ends at 01:02. In both cases, the workload duration is 1.040 h. Therefore, this scenario does not change the physical workload, the facility energy, the weighted average power, or the peak power. It only changes the electricity prices assigned to the workload samples.

Figure 7.10 shows the overview of the workload shifting scenario. The baseline facility energy remains 8.18 MWh, and the scenario facility energy is also 8.18 MWh. The weighted average power remains 7.86 MW, and the peak power remains 8.50 MW. This confirms that workload shifting does not reduce energy consumption or peak power by itself in this run.

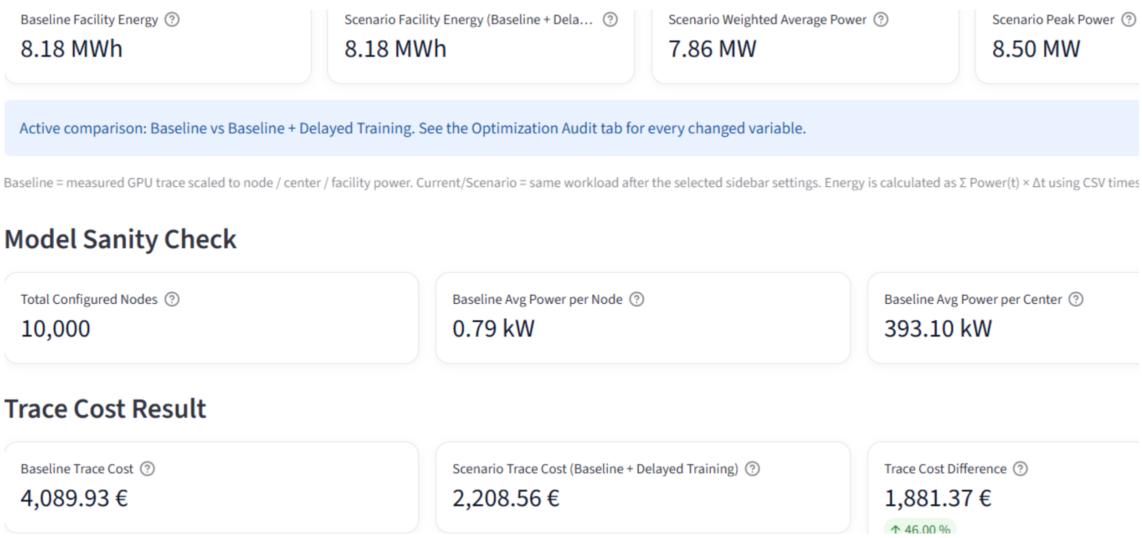


Figure 7.10.: Overview of the workload shifting scenario with unchanged energy and power values.

The cost result changes significantly. The baseline trace cost is 4,089.93 EUR, while the shifted scenario trace cost is 2,208.56 EUR. This corresponds to a cost reduction of 1,881.37 EUR, or approximately 46.00%. The reduction occurs because the same workload is moved from the evening tariff period to the night tariff period.

Figure 7.11 shows the workload timing used for the price calculation. The baseline workload runs from 18:00 to 19:02, while the shifted scenario runs from 00:00 to 01:02. The dashboard explicitly reports that the cost changes only because the workload samples fall into different tariff periods.

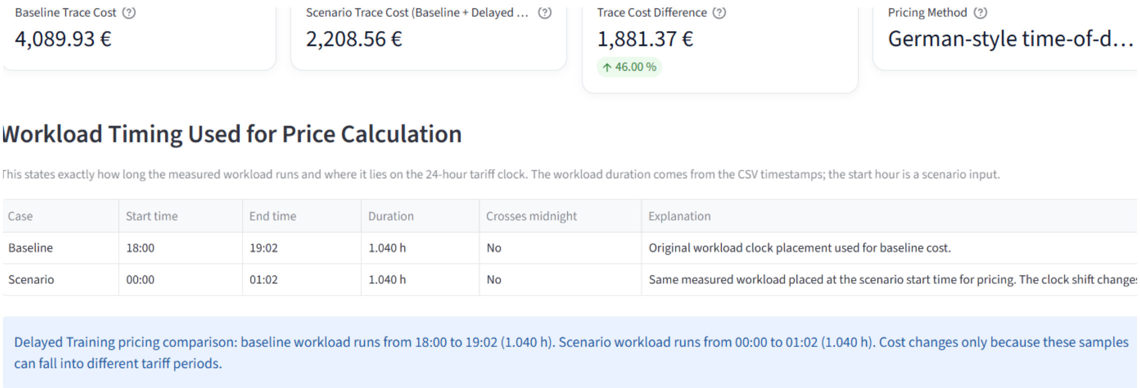


Figure 7.11.: Workload timing used for the time-dependent electricity price calculation.

Table 7.6 summarizes the comparison between the baseline timing and the shifted timing.

Table 7.6.: Cost comparison for workload shifting under time-dependent electricity prices.

Metric	**Baseline timing**	**Shifted timing**	**Change**
Start time	18:00	00:00	Shifted to night period
End time	19:02	01:02	Same duration
Trace duration	1.040 h	1.040 h	No change
Facility energy	8.18 MWh	8.18 MWh	No change
Weighted average power	7.86 MW	7.86 MW	No change
Peak power	8.50 MW	8.50 MW	No change
Trace cost	4,089.93 EUR	2,208.56 EUR	-1,881.37 EUR
Relative cost reduction	Reference	46.00% lower	Cost saving

This result shows that workload shifting is mainly a cost optimization strategy in the evaluated setup. It does not reduce the amount of electrical energy required by the workload, because the same power trace is executed for the same duration. However, it can reduce operational cost when the workload is moved from an expensive tariff period to a cheaper one.

7.3.4. Load Distribution Across Simulated HPC Centers

The load distribution scenario evaluates how limiting the number of simultaneously active simulated HPC centers affects peak demand and grid-related metrics. In this run, the scheduling strategy `rotating_centers` is used. Although 20 simulated HPC centers are available, only 10 centers are allowed to be active at any given time.

The implementation represents this scheduling decision through center-specific activity indicators. For each time step, a center is either active and receives the full center-level workload demand, or inactive and receives no workload demand. In the `rotating_centers` strategy, the active subset of centers changes over time so that the simulated work is distributed more evenly across the available centers.

Figure 7.12 shows the scheduling sanity check for this scenario. The schedule contains 707 time steps, matching the measured workload trace length. The average number of active centers is 10.00 out of 20, and both the minimum and maximum active-center count are 10. This confirms that the scheduling policy consistently limits the number of simultaneously active centers throughout the simulated trace. The per-center active share ranges from 49.5% to 50.5%, which indicates that the rotating strategy distributes activity relatively evenly across centers.

Scheduling Sanity Check

Metric	Value	How to read it
Timesteps in schedule	707	Number of measured workload samples for which a center schedule exists.
Average active centers	10.00 of 20	Mean number of centers receiving load at one timestep.
Minimum active centers	10	Lowest simultaneous center count. Should never be below 1.
Maximum active centers	10	Highest simultaneous center count. Should not exceed the sidebar maximum.
Total center-time work units	7070	Sum of active centers over all timesteps. Useful for checking whether a strategy reduces or redistributes work in t
Per-center active share range	49.5% – 50.5%	Fairness indicator. Rotating centers should be more balanced than same_centers_active.

Strategy	Max simultaneous centers	Observed max active centers	Peak power change
rotating_centers	10 / 20	10	-4.248 MW

Figure 7.12.: Scheduling sanity check for the load distribution scenario.

The dashboard reports a peak power change of −4.248 MW relative to the baseline. This reduction occurs because only half of the simulated centers receive workload demand at the same time. Since the baseline peak load was approximately 8.50 MW, the reduced peak-load snapshot of 4.25 MW is consistent with the selected limit of 10 active centers out of 20.

A key point of this scenario is the distinction between energy inside the simulated window and energy required for the same amount of work. Figure 7.13 shows that only 50.0% of the center-time work is completed inside the original simulation window. The corresponding same-work runtime is 2.08 h, while the same-work energy remains 8.18 MWh. The same-work energy saving is therefore 0.00 Wh.

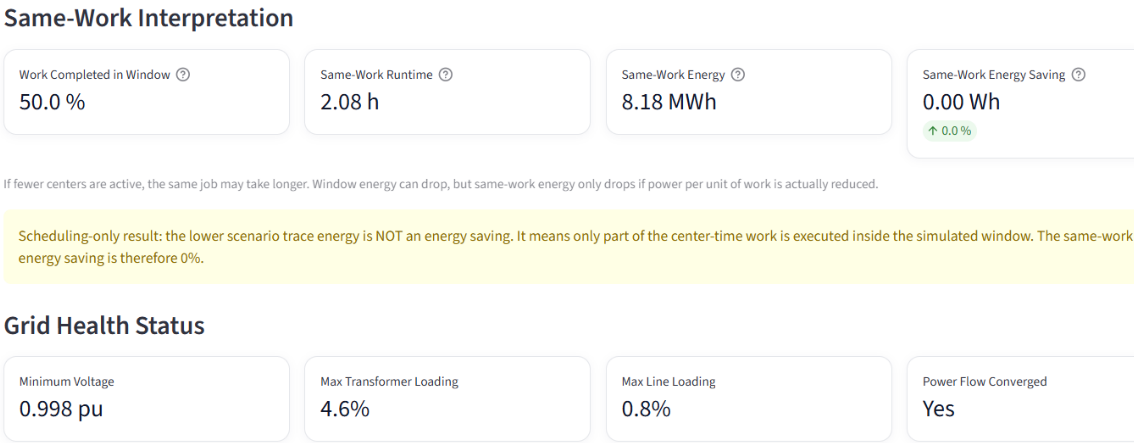
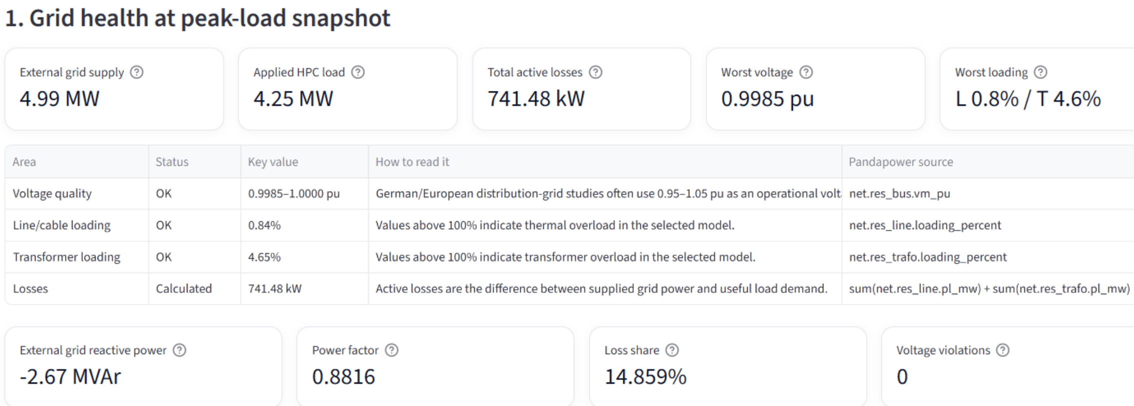


Figure 7.13.: Same-work interpretation and grid-status summary for the load distribution scenario.

This distinction is important for interpreting scheduling-based scenarios. A lower scenario energy inside a fixed simulation window should not automatically be interpreted as a true energy saving. In this implementation, load distribution reduces simultaneous activity and therefore lowers peak demand, but it does not reduce the amount of energy required to complete the same total work.

Figure 7.14 shows the grid-health indicators at the peak-load snapshot. The external grid supply is 4.99 MW, while the applied HPC load is 4.25 MW. The total active losses are reported as 741.48 kW. The worst voltage is 0.9985 pu, the maximum line loading is 0.84%, and the maximum transformer loading is 4.65%. No voltage violations are reported, and the power flow converges successfully.



Area	Status	Key value	How to read it	Pandapower source
Voltage quality	OK	0.9985–1.0000 pu	German/European distribution-grid studies often use 0.95–1.05 pu as an operational volt	net.res_bus.vm_pu
Line/cable loading	OK	0.84%	Values above 100% indicate thermal overload in the selected model.	net.res_line.loading_percent
Transformer loading	OK	4.65%	Values above 100% indicate transformer overload in the selected model.	net.res_trafo.loading_percent
Losses	Calculated	741.48 kW	Active losses are the difference between supplied grid power and useful load demand.	sum(net.res_line.pl_mw) + sum(net.res_trafo.pl_mw)

Figure 7.14.: Grid-health indicators at the peak-load snapshot for the load distribution scenario.

Table 7.7 summarizes the most important results of the load distribution scenario.

Table 7.7.: Results of the load distribution scenario.

Metric	**Value**
Scheduling strategy	`rotating_centers`
Available centers	20
Maximum simultaneously active centers	10
Observed maximum active centers	10
Average active centers	10.00 of 20
Timesteps in schedule	707
Per-center active share range	49.5%–50.5%
Peak power change	-4.248 MW
Work completed in simulated window	50.0%
Same-work runtime	2.08 h
Same-work energy	8.18 MWh
Same-work energy saving	0.00 Wh
Applied HPC load at peak snapshot	4.25 MW
External grid supply at peak snapshot	4.99 MW
Worst voltage	0.9985 pu
Maximum line loading	0.84%
Maximum transformer loading	4.65%
Voltage violations	0
Power-flow convergence	Yes

Overall, the load distribution scenario demonstrates that scheduling across multiple centers can significantly reduce simultaneous peak demand. In the evaluated setup, the peak load is nearly halved because only 10 of 20 centers are active at one time. However, this should not be interpreted as a direct energy-saving measure. The same-work interpretation shows that the total energy required for the same amount of work remains unchanged.

7.4. Capacity Analysis

The capacity analysis evaluates how the synthetic grid model behaves when the number of simulated HPC centers is increased. Unlike the previous operational scenarios, this analysis does not optimize the workload itself. Instead, it tests the grid response under increasing injected peak HPC load.

In this run, the capacity mode tests simulated center counts up to 50 HPC centers. For each tested center count, the workload peak is applied to the grid model and an AC power flow calculation is executed. The tested configuration is considered safe if the power flow converges and the selected voltage and loading limits are not violated.

Figure 7.15 shows the hosting capacity result. No failure occurs within the tested range. Therefore, the reported value of 50 centers should be interpreted as the largest safe tested value, not as an absolute physical maximum of the synthetic grid model.

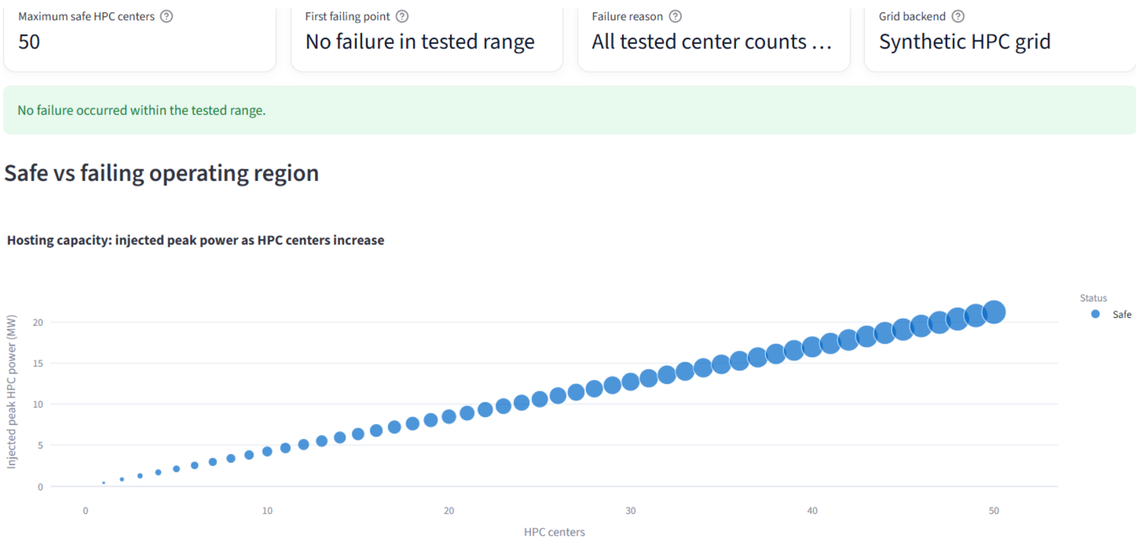


Figure 7.15.: Capacity analysis of the synthetic grid model for increasing simulated HPC center count.

The injected peak HPC power increases approximately linearly with the number of simulated centers. This is expected because the same center-level workload profile is replicated for each additional center. Since all tested points remain safe, the synthetic grid does not reach a voltage, line-loading, transformer-loading, or convergence limit within the tested range.

Table 7.8 summarizes the main capacity analysis result.

Table 7.8.: Capacity analysis result for the synthetic grid model.

Metric	**Result**
Grid backend	Synthetic HPC grid
Maximum tested HPC centers	50
Maximum safe tested HPC centers	50
First failing point	No failure in tested range
Failure reason	All tested center counts remained inside the selected limits
Status of tested points	Safe

The result indicates that the synthetic grid configuration is oversized for the tested workload range. This is consistent with the baseline grid-health results, where line and transformer loadings were far below their limits. The capacity mode therefore does not identify the real physical hosting limit of an actual HPC connection. Instead, it shows that under the selected synthetic-grid assumptions, the modeled workload can be scaled up to 50 simulated centers without violating the chosen technical limits.

7.5. Synthetic Grid and SimBench Comparison

The final evaluation compares the baseline workload on the synthetic grid model with the same workload on a SimBench-based benchmark grid. The purpose of this comparison is not to determine which grid is more realistic for a specific HPC center. Instead, it evaluates whether the workload-to-grid simulation pipeline can also be applied to a standardized benchmark network.

The same workload configuration as in the baseline run is used. The simulation contains 20 simulated HPC centers, 500 nodes per center, and 10,000 configured nodes in total. No optimization or scheduling strategy is active. Therefore, the facility-level workload values remain unchanged: the baseline facility energy is 8.18 MWh, the weighted average power is 7.86 MW, and the peak power is 8.50 MW.

Figure 7.16 shows the overview of the SimBench-based baseline run. Since the workload and scaling assumptions are unchanged, the energy, average power, peak power, and cost values remain identical to the synthetic-grid baseline. The difference between the two runs is therefore not the workload itself, but the grid model in which the workload-derived load profile is evaluated.

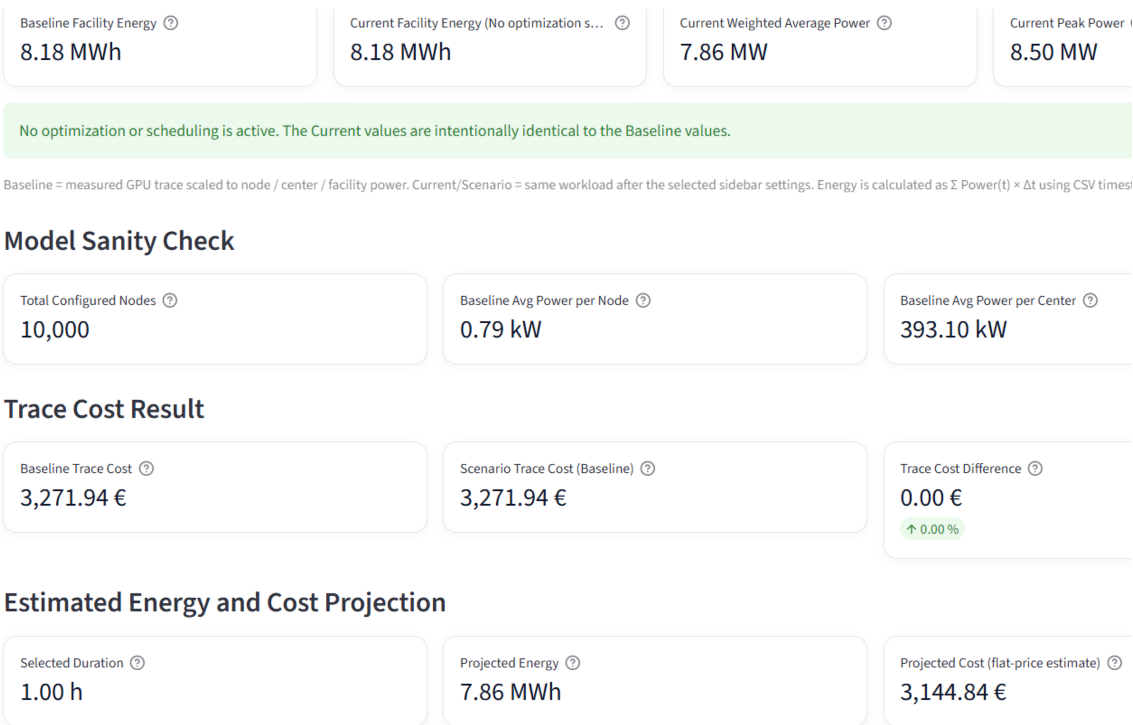
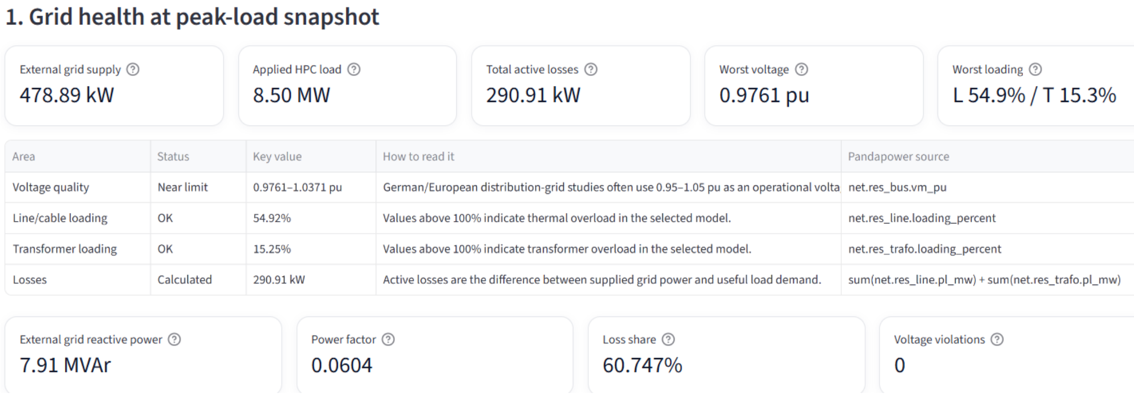


Figure 7.16.: Overview of the baseline workload evaluated with the SimBench-based grid model.

Figure 7.17 shows the grid-health indicators for the SimBench-based run. At the peak-load snapshot, the applied HPC load is 8.50 MW. The worst voltage is 0.9761 pu, which remains above the selected lower voltage limit of 0.95 pu but is noticeably lower than in the synthetic-grid baseline. The maximum line loading is 54.92%, and the maximum transformer loading is 15.25%. No voltage violations are reported, and the power flow converges successfully.



Area	Status	Key value	How to read it	Pandapower source
Voltage quality	Near limit	0.9761–1.0371 pu	German/European distribution-grid studies often use 0.95–1.05 pu as an operational volta	net.res_bus.vm_pu
Line/cable loading	OK	54.92%	Values above 100% indicate thermal overload in the selected model.	net.res_line.loading_percent
Transformer loading	OK	15.25%	Values above 100% indicate transformer overload in the selected model.	net.res_trafo.loading_percent
Losses	Calculated	290.91 kW	Active losses are the difference between supplied grid power and useful load demand.	sum(net.res_line.pl_mw) + sum(net.res_trafo.pl_mw)

Figure 7.17.: Grid-health indicators for the SimBench-based baseline simulation.

The comparison shows that the same workload can lead to different grid-related results depending on the grid model. In the synthetic-grid baseline, the worst voltage was 0.9985 pu, the maximum line loading was approximately 0.84%, and the maximum transformer loading was approximately 4.65%. In the SimBench-based run, the worst voltage decreases to 0.9761 pu, while the maximum line loading increases to 54.92%. This indicates that the SimBench benchmark grid is more sensitive to the additional HPC loads under the selected connection assumptions.

Figure 7.18 shows the SimBench-based pandapower topology used in this evaluation. In contrast to the synthetic grid, the benchmark grid has an existing network structure with predefined buses, lines, transformers, and operating conditions. The simulated HPC loads are inserted into this benchmark structure as additional controllable loads.

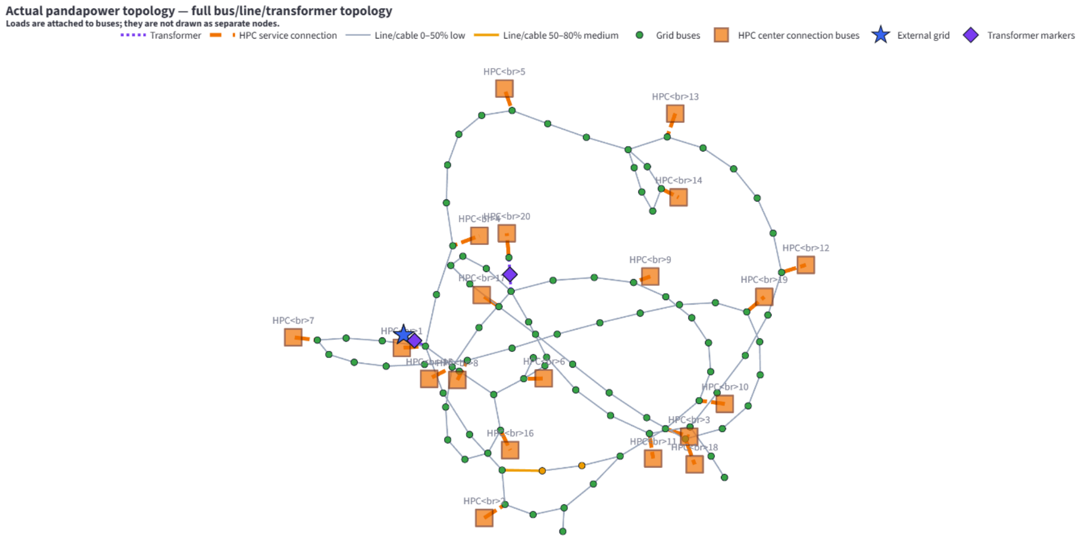


Figure 7.18.: SimBench-based pandapower topology with inserted simulated HPC connection buses.

Table 7.9 summarizes the main comparison between the synthetic baseline and the SimBench-based baseline.

Table 7.9.: Comparison between synthetic-grid and SimBench-based baseline results.

Metric	**Synthetic grid**	**SimBench-based grid**
Facility energy	8.18 MWh	8.18 MWh
Weighted average power	7.86 MW	7.86 MW
Peak power	8.50 MW	8.50 MW
Applied HPC load at peak	8.50 MW	8.50 MW
Worst voltage	0.9985 pu	0.9761 pu
Maximum line loading	0.84%	54.92%
Maximum transformer loading	4.65%	15.25%
Voltage violations	0	0
Power-flow convergence	Yes	Yes

The external grid supply, power factor, and loss share reported for the SimBench-based case should be interpreted carefully, because the benchmark grid may already contain existing loads, generation, and internal power exchanges. These values are therefore not directly comparable to the synthetic-grid case. For this comparison, the most relevant indicators are voltage behavior, component loading, violation counts, and convergence.

Overall, the SimBench-based evaluation confirms that the workload-to-grid simulation pipeline can be applied not only to the synthetic grid model but also to a standardized benchmark grid. The results also show that grid topology and existing network conditions strongly influence the observed grid impact. While the synthetic grid remains almost unaffected by the baseline load, the SimBench-based grid shows a clearer voltage drop and substantially higher line loading, although all values remain within the selected limits.

7.6. Summary of Results

The results show that the developed simulation framework can transform measured single-GPU workload traces into scaled HPC-center load profiles and evaluate their effects in power system simulations. The baseline run provides a reference case in which 20 simulated HPC centers with 500 nodes each are evaluated on the synthetic grid model. In this configuration, the baseline facility energy is 8.18 MWh, the weighted average power is 7.86 MW, and the peak power is 8.50 MW. The synthetic grid remains within the selected operating limits, with no voltage violations, line overloads, or transformer overloads.

The GPU power limiting scenario reduces the GPU power component by applying a power factor of 0.90. This decreases peak power from 8.50 MW to 7.92 MW and reduces the weighted average power from 7.86 MW to 7.35 MW. However, because the utilization-based runtime model increases the trace duration from 1.040 h to 1.091 h, the integrated energy decreases only from 8.18 MWh to 8.02 MWh. This shows that GPU power limiting is more effective at reducing instantaneous power and peak load than at reducing total energy when runtime effects are considered.

The PUE-based facility efficiency scenario produces a different effect. By reducing the facility overhead, the scenario decreases facility energy from 8.18 MWh to 7.90 MWh without changing workload runtime. The weighted average power decreases to 7.59 MW, and the peak power decreases to 8.20 MW. This confirms that PUE-based improvements affect facility-level energy and power directly, while leaving the computational workload unchanged.

The workload shifting scenario demonstrates the difference between energy consumption and operational cost. The workload energy and power remain unchanged, because the same measured workload profile is executed for the same duration. However, shifting the workload from 18:00 to 00:00 reduces the trace cost from 4,089.93 EUR to 2,208.56 EUR under the selected time-dependent electricity price model. This corresponds to a cost reduction of 1,881.37 EUR. The result shows that workload shifting is mainly a cost optimization strategy in this setup, not an energy-saving strategy.

The load distribution scenario shows that limiting simultaneous activity across centers can reduce peak demand. With the `rotating_centers` strategy, only 10 of 20 simulated centers are active at each time step. This reduces the peak power by approximately 4.248 MW. However, the same-work interpretation shows that only 50.0% of the work is completed within the original simulation window, while the same-work energy remains 8.18 MWh. Therefore, the scenario reduces simultaneous load but does not reduce the total energy required to complete the same amount of work.

The capacity analysis tests the synthetic grid for increasing numbers of simulated HPC centers. No failure occurs within the tested range up to 50 centers. This means that 50 centers should be interpreted as the largest safe tested value, not as an absolute physical hosting limit. The result is consistent with the baseline grid-health results, where the synthetic grid showed low line and transformer loading.

The SimBench comparison shows that the same workload-derived load profile can produce different grid-related effects depending on the grid model. While the synthetic-grid baseline shows a worst voltage of 0.9985 pu and very low component loading, the SimBench-based run shows a worst voltage of 0.9761 pu and a maximum line loading of 54.92%. Both simulations remain within the selected limits, but the SimBench-based grid is more sensitive to the inserted HPC loads.

Overall, the results demonstrate that time-dependent workload modeling provides more insight than a static average-power assumption. Energy consumption, peak demand, operational cost, voltage behavior, and component loading react differently to the evaluated scenarios. The interpretation of each scenario depends on its mechanism: GPU power limiting changes workload power and runtime, PUE improvements change facility overhead, workload shifting changes tariff placement, and load distribution changes simultaneous activity. These distinctions are important when evaluating ML workload operation in HPC-like environments.

8. Discussion

This chapter discusses the results of the simulation framework and interprets their meaning in relation to the thesis objective. The focus is on what the simulation results show, how the operational scenarios should be interpreted, and which limitations must be considered when evaluating the approach.

8.1. Interpretation of the Baseline Results

The baseline simulation demonstrates that the developed pipeline can transform measured single-GPU workload traces into scaled HPC-center load profiles and evaluate them in a pandapower-based grid model. The baseline configuration uses 20 simulated HPC centers with 500 nodes per center. Under these assumptions, the framework produces a facility-level load profile with a peak demand of 8.50 MW and a trace energy of 8.18 MWh.

In the synthetic grid model, this load does not lead to critical grid behavior. The voltage remains close to nominal, and line and transformer loading remain far below their limits. This indicates that the selected synthetic grid is strong enough for the evaluated baseline load. The result also confirms that the workload-to-grid pipeline is numerically stable for the chosen configuration.

However, the baseline result should not be interpreted as a statement about the real hosting capacity of German HPC infrastructure. The synthetic grid is an assumption-based model with representative voltage levels and simplified component sizing. Therefore, the baseline mainly shows that the simulation framework works consistently under the selected assumptions. It does not prove that a real HPC center with the same load would have the same grid behavior.

8.2. Interpretation of Operational Strategies

The evaluated scenarios show that different operational strategies affect different parts of the system. This is one of the central findings of the thesis. A strategy that reduces peak power does not necessarily reduce total energy, and a strategy that reduces cost does not necessarily reduce physical electricity consumption.

The GPU power limiting scenario reduces instantaneous demand by lowering the GPU power component. This leads to lower average power and lower peak power. However, the modeled runtime increases, which partly compensates for the power reduction. Therefore, the energy reduction is smaller than the reduction in GPU power. This confirms that GPU power limiting must be evaluated together with performance effects. Considering only the lower power level would overestimate the energy benefit.

The PUE-based facility efficiency scenario has a different mechanism. It reduces facility overhead while leaving the workload itself unchanged. As a result, the workload runtime does not increase. The energy and peak-power reductions are therefore easier to interpret than in the GPU power limiting case. However, the reduction is smaller than the cooling overhead factor itself because the factor is applied only to the non-IT overhead part of the facility demand, not to the entire load.

The workload shifting scenario shows that energy consumption and operational cost must be evaluated separately. The same workload profile can consume the same amount of energy but lead to different costs if it is executed during a different tariff period. In the evaluated case, shifting the workload to the night period reduces cost without changing the physical energy demand. This means that workload shifting is mainly a cost optimization strategy in this setup, not an energy-saving strategy.

The load distribution scenario mainly affects simultaneity. By allowing only 10 of 20 simulated centers to be active at each time step, the peak load is reduced. However, the same-work interpretation shows that this does not reduce the total energy required to complete the same amount of work. Instead, only part of the work is completed inside the original simulation window. This distinction is important because otherwise scheduling-based scenarios can appear to save energy simply by moving work outside the evaluated time interval.

Overall, the scenario results show that operational strategies must be interpreted according to their mechanism. GPU power limiting changes workload power and runtime, PUE improvements change facility overhead, workload shifting changes tariff placement, and load distribution changes simultaneous activity. These mechanisms are not interchangeable.

8.3. Influence of the Grid Model

The comparison between the synthetic grid and the SimBench-based grid shows that the grid model strongly influences the observed grid-related results. The same workload-derived load profile produces very low component loading in the synthetic grid, while the SimBench-based grid shows a stronger voltage drop and substantially higher line loading.

This does not mean that one grid model is generally correct and the other is wrong. The two models serve different purposes. The synthetic grid provides a transparent and controllable environment in which voltage levels, connection points, and load scaling can be defined directly. This is useful for controlled scenario analysis. The SimBench-based grid provides a standardized benchmark structure and therefore helps test whether the same workload-to-grid pipeline can also be applied in a less manually constructed grid environment.

The difference between the two grid results also shows why grid topology matters. Voltage behavior and component loading depend not only on the total amount of load, but also on line impedances, transformer ratings, existing loads, generation, and the selected connection points. Therefore, a workload profile cannot be evaluated independently of the network into which it is inserted.

Some dashboard indicators in the SimBench case, such as external grid supply, power factor, and loss share, must be interpreted carefully. Since the SimBench network can contain existing loads, generation, and internal power exchanges, these values are not directly comparable to the synthetic-grid case. For this reason, the comparison in this thesis focuses mainly on voltage behavior, line loading, transformer loading, violation counts, and power-flow convergence.

8.4. Usefulness of the Simulation Framework

The simulation framework is useful because it connects workload-level power behavior with grid-level evaluation. Instead of using only nominal GPU power or a static average value, the framework preserves the time-dependent behavior of the measured workload traces. This makes it possible to evaluate peak demand, accumulated energy, operational cost, voltage behavior, transformer loading, and line loading in one workflow.

A strength of the framework is its modular structure. Workload preprocessing, power modeling, grid construction, simulation execution, scenario handling, cost estimation, and capacity analysis are separated into different components. This makes the approach reproducible and extendable. For example, newer workload traces can replace the current CSV files without changing the grid simulation logic. Similarly, a more detailed grid model could replace the synthetic grid while keeping the workload modeling pipeline.

The framework is especially useful for comparative analysis. The absolute results depend strongly on the selected assumptions, such as number of centers, nodes per center, PUE, non-GPU node power, and grid model. However, comparing scenarios against the same baseline provides insight into how different operational decisions affect the results. This makes the framework suitable for exploring trade-offs between energy, peak power, cost, and grid-related behavior.

Another useful aspect is the explicit treatment of same-work interpretation. The load distribution scenario shows that a lower energy value inside a fixed simulation window is not automatically a true energy saving. By distinguishing between simulated-window results and same-work results, the framework avoids misleading conclusions about scheduling-based strategies.

8.5. Limitations

The main limitation of this thesis is the limited availability of detailed real-world HPC infrastructure data. The available traces represent controlled single-GPU workload executions. They are not complete node-level, rack-level, center-level, or facility-level measurements. Therefore, the load profiles used in the simulation are estimated by scaling the measured GPU traces with additional assumptions.

The non-GPU node power components are modeled as constant values. In reality, CPU, memory, storage, and network power also change over time depending on workload behavior and system activity. The current implementation simplifies these components because detailed component-level measurements were not available. This makes the model transparent, but it also limits the accuracy of the node-level power estimate.

Facility overhead is represented through PUE. This is appropriate for aggregated facility-level estimation, but it does not model cooling systems, power distribution losses, airflow, thermal dynamics, or uninterruptible power supply behavior in detail. Therefore, PUE-based results should be interpreted as simplified facility-level scaling estimates rather than detailed data center engineering results.

The synthetic grid model is also simplified. It uses representative voltage levels and a controlled topology, but it does not reproduce a specific German distribution grid or a specific HPC facility connection. The SimBench-based grid provides a standardized benchmark network, but it is also not a real measured HPC site. As a result, the grid-related results are scenario-based estimates and should not be interpreted as direct operational statements about an existing facility.

Reactive power is approximated using a fixed proportional assumption. This allows AC power flow simulations to be executed, but it does not capture detailed power factor behavior of real HPC equipment. Actual power factor behavior can depend on power supplies, converters, compensation equipment, and operating conditions.

The cost model is simplified as well. It uses time-dependent electricity prices to estimate operational cost, but it does not include all components of real electricity billing. Grid fees, taxes, demand charges, procurement contracts, balancing costs, and other market mechanisms are not modeled. Therefore, the cost results are useful for comparing workload timing scenarios, but they are not a complete electricity billing model.

8.6. Validity of the Results

The internal validity of the results is supported by the consistency of the implemented scaling and simulation pipeline. The baseline values are internally coherent: average power, node count, center count, and center-level power are consistent with each other. The scenario results also behave according to the expected mechanisms. GPU power limiting reduces peak power while increasing runtime, PUE efficiency reduces facility overhead without runtime change, workload shifting changes cost without changing energy, and load distribution reduces simultaneity without reducing same-work energy.

The external validity is more limited. The results cannot be generalized directly to real German HPC centers or real distribution grids. The workload traces, non-GPU power assumptions, facility overhead, and grid models are simplified. Therefore, the results should be interpreted as scenario-based simulation outcomes rather than measured real-world behavior.

The reproducibility of the framework is a strength. The workload traces are stored as CSV files, the simulation parameters are configurable, and the grid models are generated through code. This means that the same assumptions can be rerun and inspected. If more detailed data becomes available, the current assumptions can be replaced step by step.

8.7. Implications for Future HPC Operation

The results suggest that future HPC and ML workload operation should consider not only total energy consumption, but also when and where power demand occurs. Peak demand, runtime effects, facility overhead, electricity price timing, and grid topology can all influence the operational outcome.

For ML workloads, power management strategies should be evaluated together with performance effects. A lower GPU power limit may reduce instantaneous power and peak load, but if runtime increases, the energy benefit may be smaller than expected. Facility efficiency improvements can reduce total facility energy without changing workload runtime, but they depend on infrastructure-level improvements.

Scheduling and load distribution can help reduce simultaneous demand, but they must be evaluated on a same-work basis. Otherwise, a scenario may appear to reduce energy simply because part of the workload has been delayed beyond the evaluated time window. This is especially important when comparing operational strategies in simulation.

Overall, the results support the idea that workload-aware and grid-aware analysis can help evaluate future HPC operation. Even with simplified assumptions, the framework makes visible how different operational decisions affect energy, peak power, cost, and grid-related metrics.

9. Conclusion

This thesis developed a simulation framework for analyzing the power demand of machine learning workloads in HPC-like environments using pandapower. The framework connects measured single-GPU workload traces with power system simulation by transforming GPU power measurements into time-dependent HPC-center load profiles. These profiles are scaled using configurable infrastructure assumptions and evaluated in both a synthetic grid model and a SimBench-based benchmark grid.

The main objective was to study how ML workload power behavior affects energy consumption, peak power demand, operational cost, and grid-related quantities such as voltage behavior, transformer loading, and line loading. Since detailed real-world HPC infrastructure data was only partially available, the thesis used representative modeling assumptions instead of attempting to reproduce one specific real HPC facility.

The results show that time-dependent workload modeling provides more insight than static average-power assumptions. The baseline evaluation demonstrates that measured GPU traces can be scaled to a multi-center HPC scenario and used as input for AC power flow simulations. Under the selected baseline configuration, the synthetic grid remained within the chosen voltage and loading limits.

The evaluated scenarios show that different operational strategies affect the system in different ways. GPU power limiting reduced peak demand and average power, but the energy reduction was smaller because the modeled runtime increased. PUE-based facility efficiency reduced facility-level energy and cost without changing workload runtime. Workload shifting did not reduce physical energy consumption, but it reduced operational cost under the selected time-dependent price assumptions. Load distribution reduced simultaneous peak demand, but the same-work interpretation showed that it did not reduce the total energy required to complete the same amount of work.

The capacity analysis showed no failure within the tested range of up to 50 simulated HPC centers in the synthetic grid model. This result must be interpreted as the largest safe tested value under the selected assumptions, not as an absolute physical hosting limit. The SimBench comparison showed that the same workload-derived load profile can lead to different grid-related behavior depending on the grid model. In particular, the SimBench-based grid showed a stronger voltage drop and higher line loading than the synthetic grid, while still remaining within the selected limits.

Overall, the thesis demonstrates that workload-derived power traces can be integrated into power flow simulations in a reproducible way. The developed framework makes it possible to evaluate energy, peak power, cost, voltage behavior, transformer loading, and line loading together. It also shows that operational strategies must be interpreted according to their mechanism: GPU power limiting affects workload power and runtime, PUE changes affect facility overhead, workload shifting affects tariff placement, and load distribution affects simultaneous activity.

The main limitation of the work is that the model is based on single-GPU traces and representative assumptions rather than detailed real-world HPC infrastructure measurements. Future work could improve the framework by integrating larger multi-GPU traces, measured node-level and facility-level power data, more detailed cooling and power distribution models, and site-specific grid connection data. Further extensions could also include improved reactive power modeling, more realistic electricity tariff structures, and validation against measured data from operational HPC facilities.

Despite these limitations, the framework provides a useful basis for studying the interaction between ML workloads and electrical grid models. It supports scenario-based analysis and can be extended as more detailed workload and infrastructure data becomes available.

Appendix A.

Simulation Configuration

This appendix summarizes the main configuration values used for the simulation runs presented in Chapter 7. The purpose is to make the evaluation settings transparent and reproducible.

A.1. Baseline Configuration

Table A.1.: Baseline simulation configuration used for the main evaluation.

Parameter	**Value**
Workload type	Training workload
Input data	CSV-based single-GPU power traces
Number of raw workload runs	4
Representative samples	707
Average GPU power	399.43 W
Average GPU utilization	67.2%
Simulated HPC centers	20
Nodes per center	500
Total configured nodes	10,000
Clusters per center	4
Racks per cluster	10
Facility overhead	Modeled through PUE
PUE value	1.30
CPU power per node	150 W
Grid backend	Synthetic HPC grid
Simulation mode	AC power flow
Scenario	Baseline without optimization
Scheduling	Disabled

A.2. Scenario Configuration

Table A.2.: Scenario-specific configuration used in the evaluation.

Scenario	Configuration
Baseline	Original workload-derived load profile without optimization or scheduling
GPU power limiting	GPU power factor of 0.90 with utilization-based runtime slowdown enabled
PUE-based facility efficiency	Cooling overhead factor of 0.85; effective PUE reduced from 1.300 to 1.255
Workload shifting	Baseline workload start at 18:00; shifted workload start at 00:00
Load distribution	20 available centers; 10 centers active at each time step; strategy: `rotating_centers`
Capacity analysis	Synthetic grid tested up to 50 simulated HPC centers
SimBench comparison	Baseline workload evaluated on a SimBench-based benchmark grid

A.3. Capacity Analysis Settings

Table A.3.: Capacity analysis settings.

Parameter	Value
Grid backend	Synthetic HPC grid
Maximum tested HPC centers	50
Minimum voltage limit	0.95 pu
Maximum loading limit	100%
Failure criteria	Power-flow non-convergence, voltage below limit, line overload, or transformer overload
Result interpretation	Largest safe tested value, not absolute physical grid capacity

A.4. Electricity Price Assumptions

Table A.4.: Time-dependent electricity price assumptions used for cost evaluation.

Time period	Electricity price
00:00–06:00	0.27 EUR/kWh
06:00–10:00	0.44 EUR/kWh
10:00–16:00	0.31 EUR/kWh
16:00–21:00	0.50 EUR/kWh
21:00–24:00	0.35 EUR/kWh

A.5. SimBench Configuration

Table A.5.: SimBench configuration used for the benchmark-grid comparison.

Parameter	**Value**
Grid backend	SimBench-based benchmark grid
SimBench code	`1-MV-rural-0-sw`
Existing SimBench loads/generation	Enabled
Added loads	Simulated HPC-center loads
Workload configuration	Same as baseline configuration
Scenario	Baseline without optimization

Appendix B.

Input Data and Code Structure

This appendix summarizes the input data structure and the main implementation files used in the simulation framework.

B.1. Input CSV Structure

The workload input data consists of CSV-based traces from controlled single-GPU workload executions. The columns used by the simulation framework are summarized in Table B.1.

Table B.1.: Typical CSV columns used in the workload traces.

Column	Description
`timestamp`	Measurement timestamp used for time-series alignment
`epoch`	Training epoch, if available
`step`	Workload step, if available
`gpu_util_percent`	GPU utilization in percent
`gpu_power_w`	Measured GPU power in watts
`num_gpus`	Number of GPUs used during the run
`total_gpu_power_w`	Total GPU power in watts
`cpu_util_percent`	CPU utilization in percent
`note`	Additional note about the measurement

B.2. Main Implementation Files

The simulation framework is organized into a dashboard file and several Python modules. The main files are summarized in Table B.2.

Table B.2.: Main implementation files of the simulation framework.

File	Purpose
app.py	Streamlit dashboard for configuration, simulation execution, and result visualization
profile _builder.py	Loads, aligns, and averages CSV-based GPU power traces
power_model.py	Converts workload-level GPU power into node-level, center-level, and facility-level power
grid_model.py	Builds the synthetic grid and integrates the SimBench-based grid model
run _simulation.py	Executes time-series pandapower simulations and extracts grid metrics
optimization _scenarios.py	Implements GPU power limiting, PUE changes, workload shifting, and load distribution
cost_model.py	Estimates operational cost using time-dependent electricity prices
capacity _analysis.py	Performs capacity analysis for increasing simulated HPC center counts
energy _projection.py	Calculates energy and cost projections from simulated power demand

B.3. Workload-to-Grid Processing Chain

The implemented processing chain converts the workload measurements into grid-simulation inputs. The main steps are summarized in Table B.3.

Table B.3.: Processing chain from workload trace to grid simulation.

Step	Description
Trace loading	CSV workload traces are loaded from the training or inference data directory
Preprocessing	Timestamps are sorted, sampling intervals are calculated, and repeated runs are aligned
Workload profile	GPU power values are averaged into a representative time-series profile
Node scaling	GPU power is combined with assumed non-GPU node power components
Center scaling	Node power is multiplied by the number of nodes per simulated HPC center
Facility scaling	IT power is multiplied by PUE to estimate facility-level power demand
Grid assignment	Facility power is converted to MW and assigned to pandapower load elements
Power flow	AC power flow is executed for each time step or selected peak-load snapshot
Evaluation	Energy, peak power, voltage, loading, convergence, and cost metrics are extracted

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Selbstständigkeitserklärung

Hiermit erkläre ich, dass ich die vorliegende Masterarbeit selbstständig und nur unter Verwendung der angegebenen Quellen und Hilfsmittel verfasst habe. Alle Stellen, die wörtlich oder sinngemäß aus anderen Werken entnommen wurden, sind als solche kenntlich gemacht. Die Arbeit wurde in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegt.

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